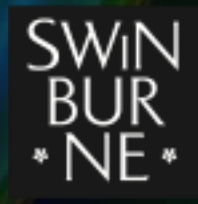


Investigating the Formation of Heavy Double Neutron Stars

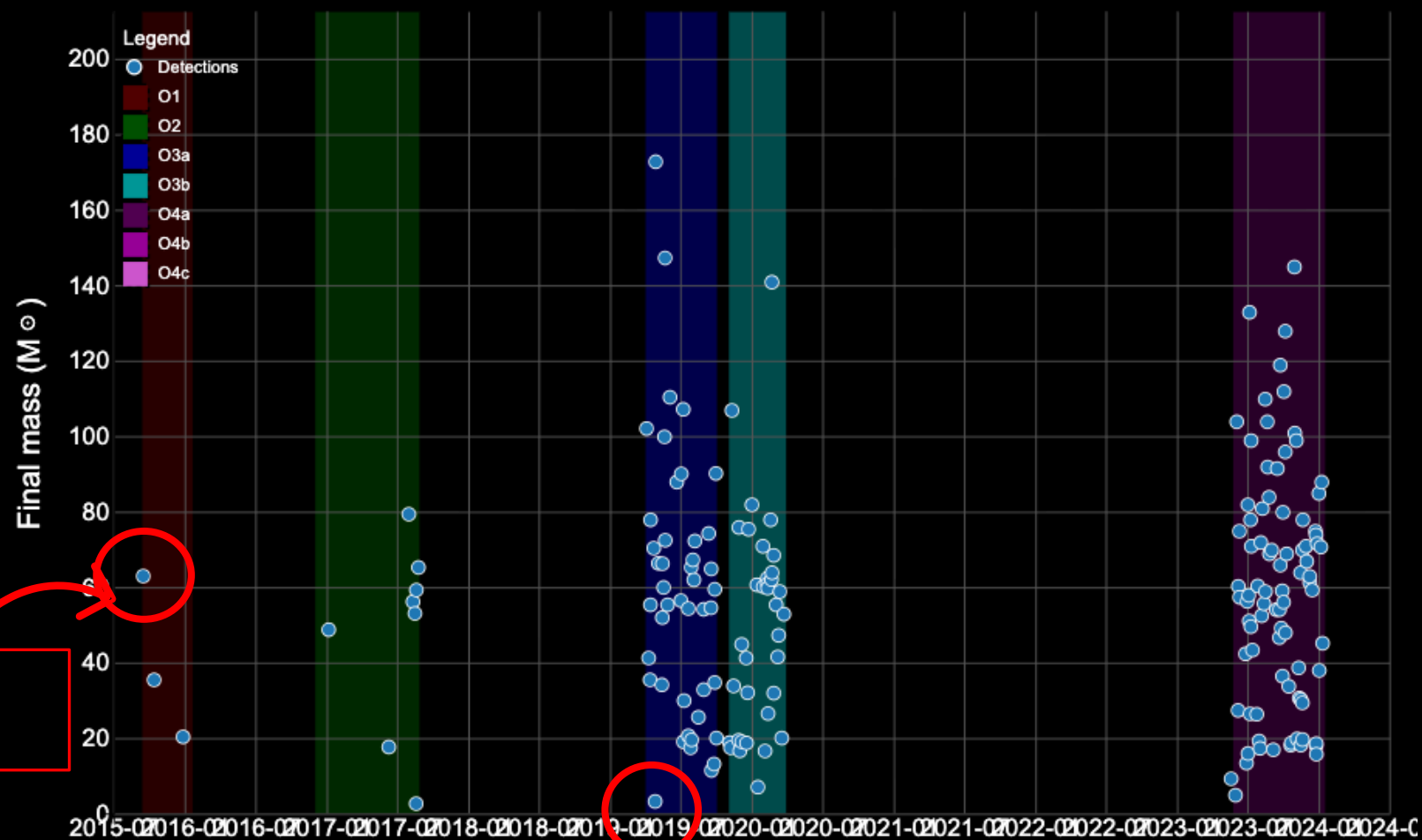
Ashwathi Nair

Supervisors : Dr. Simon Stevenson, Prof. Jarrod Hurley



Centre for
Astrophysics and
Supercomputing

LIGO-Virgo-KAGRA Compact Binary Catalogue

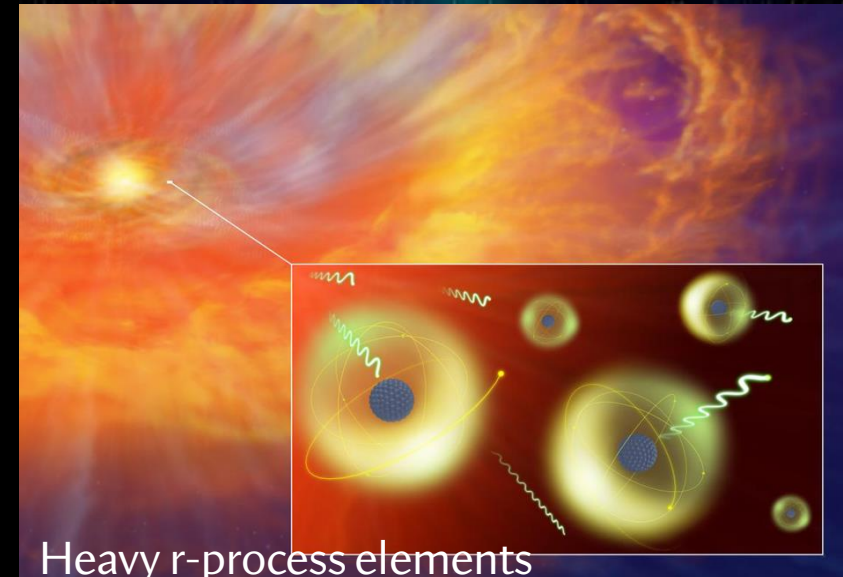
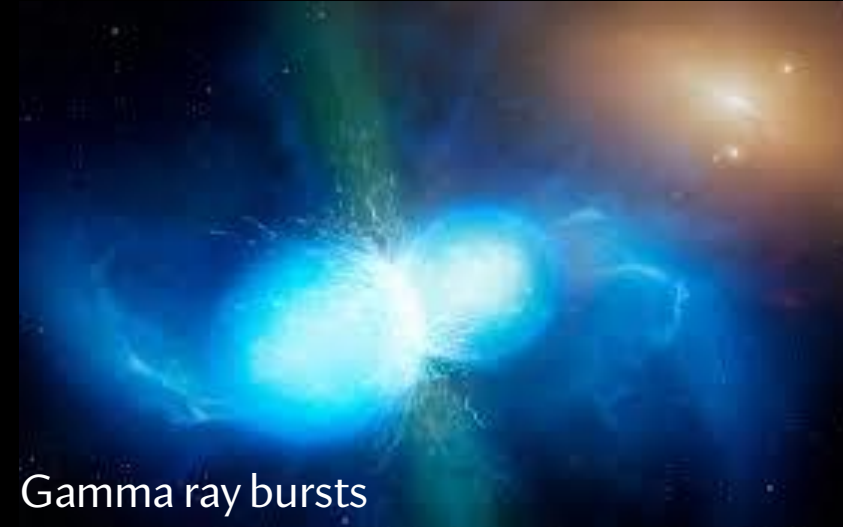


Binary Black Hole
GW150914

Double Neutron Star (DNS)
GW190425

credit: cardiffgravity.org

Why care about double neutron stars?



Radio Telescopes



Parkes Radio Observatory

~ 26 Galactic DNS in the field
Total mass: $2.5\text{--}2.9 M_{\odot}$

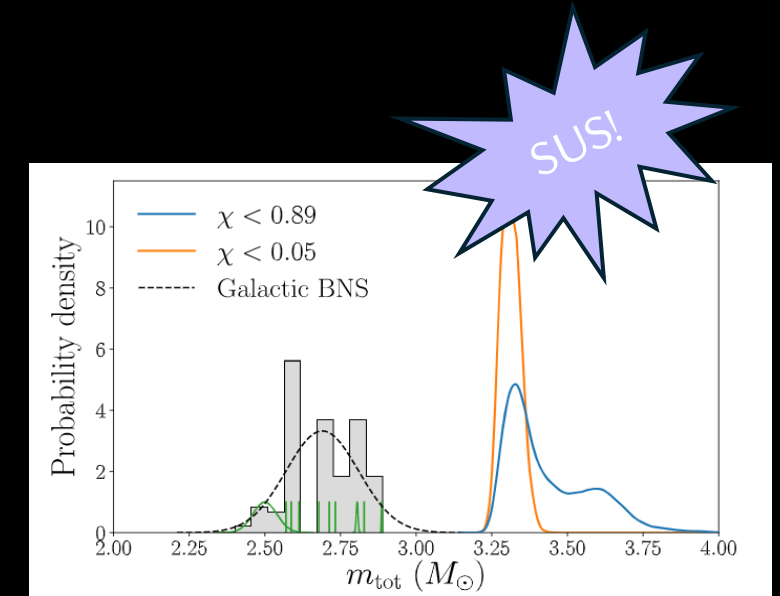
Gravitational Wave Observatories



LIGO Livingston

GW170817: Total mass: $\sim 2.8 M_{\odot}$
GW190425: Total mass: $\sim 3.4 M_{\odot}$

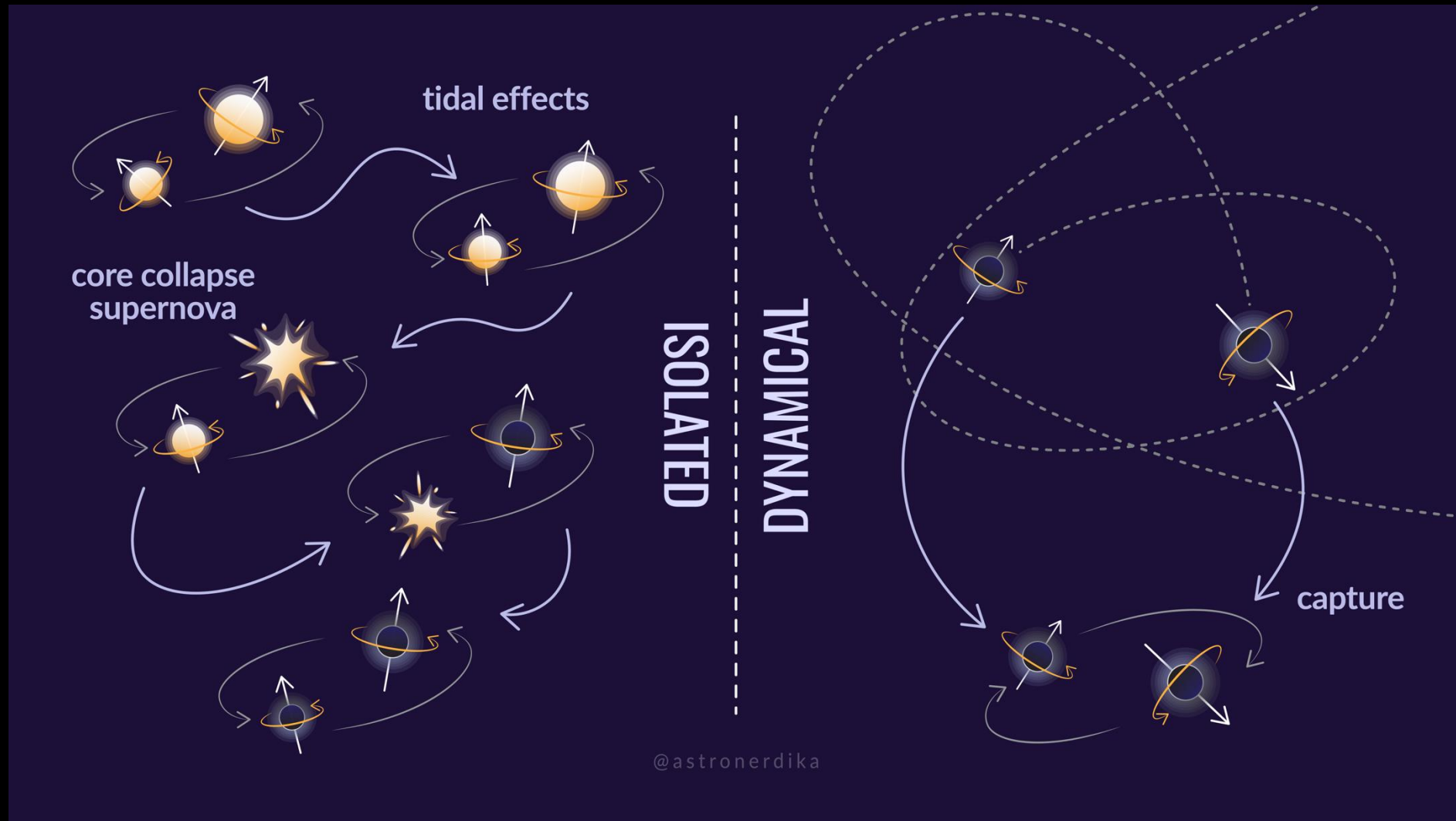
GW231109_235456 : $\sim 2.95 M_{\odot}$ (if astrophysical!)



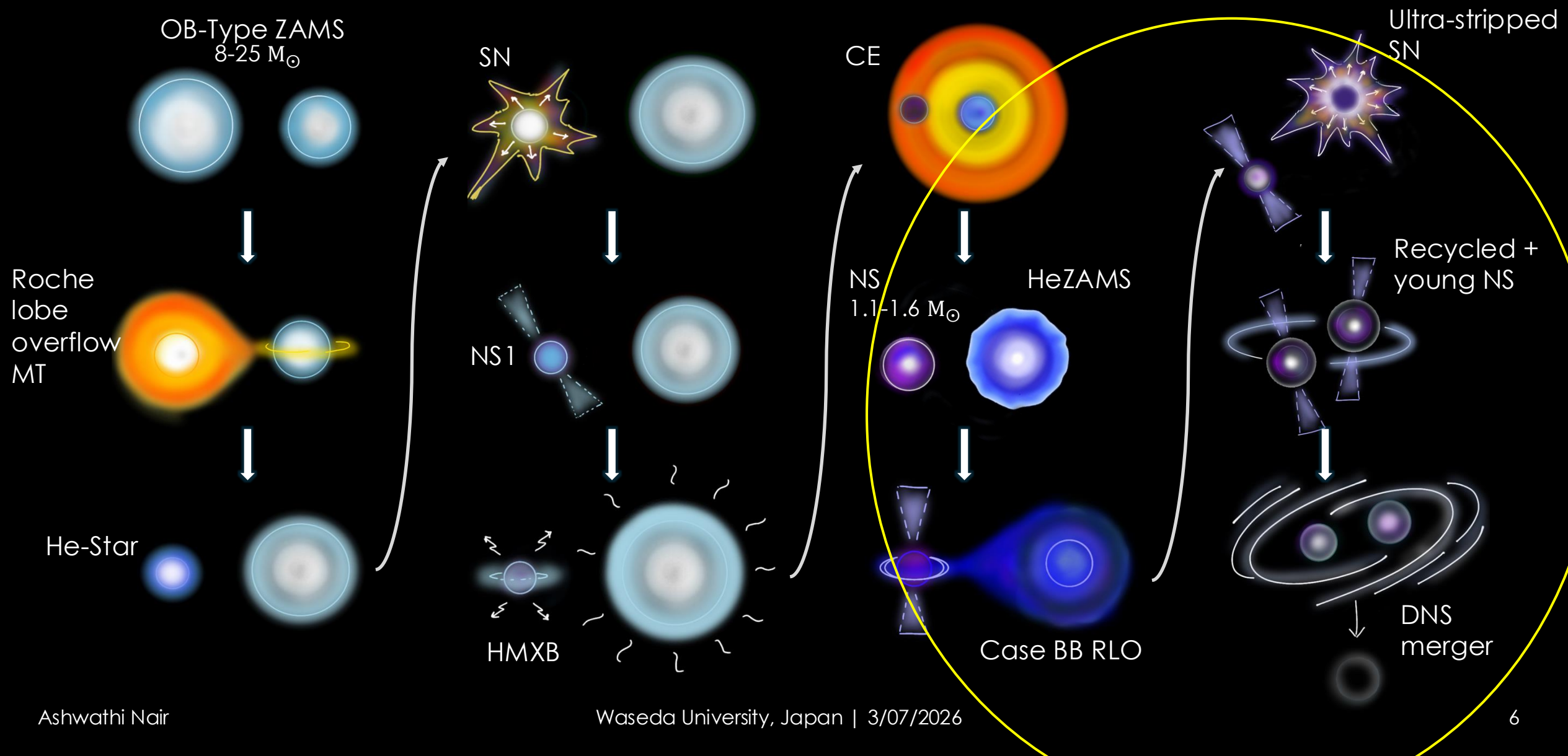
Abbot et al. 2020

DNS Total mass $\geq 3 M_{\odot}$: Heavy DNSs

Formation Channels of Compact Binary Mergers

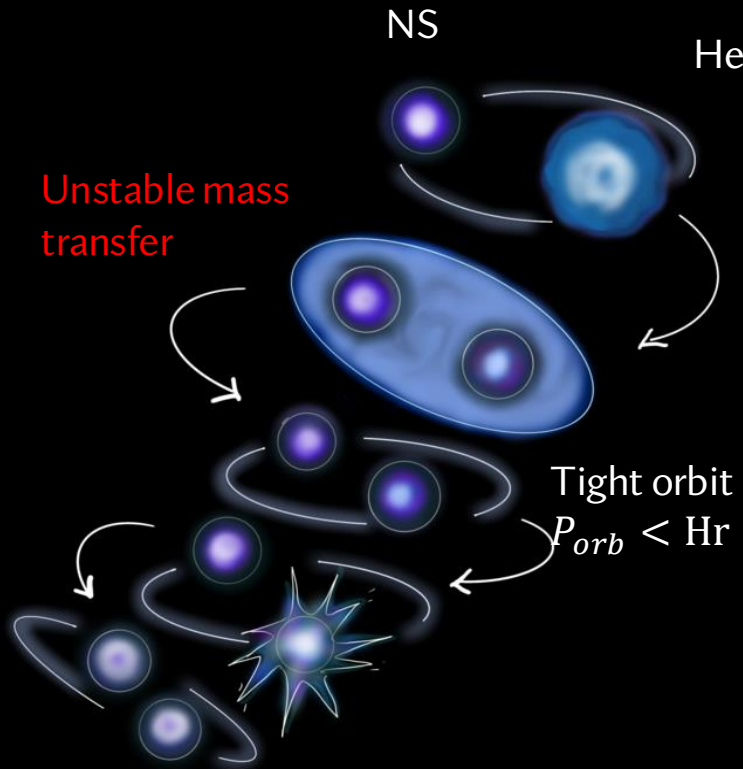


Standard Double Neutron Star Formation Pathway at $Z_{\odot}$ – T. Tauris et al. 2017



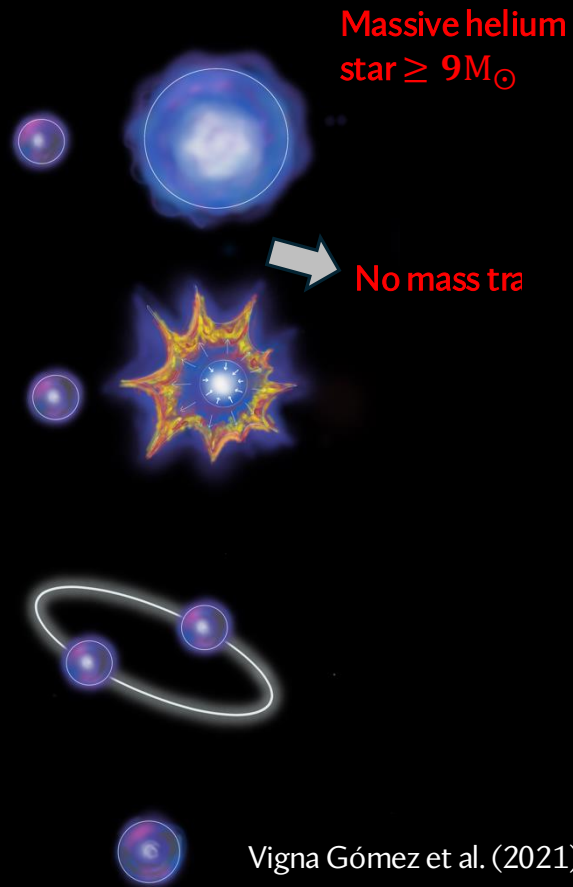
Alternate formation channels/pathways

Fast Merger



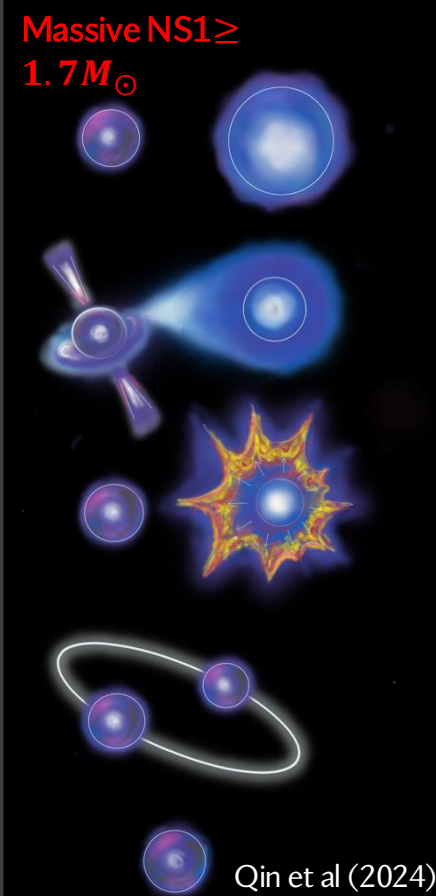
B.P. Abbott et al. (2020),
Romero

Massive He star



Vigna Gómez et al. (2021)

Massive NS1



Low metallicity

(Same as standard channel)

?

Giacobbo & Mapelli (2018)

Basically, similar formation pathway just different initial conditions and following mass transfer processes!

No self-consistent, unbiased formation pathway to explain heavy double neutron star formation!

My role : Test all the formation channels and find the winner!

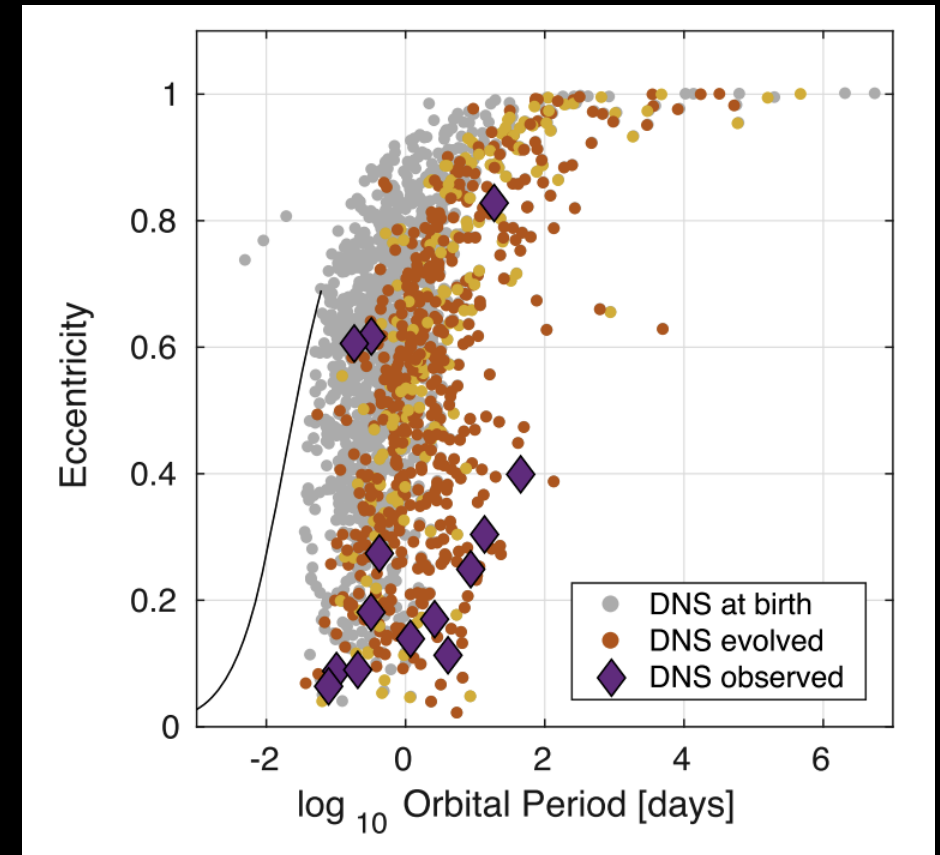
Common approach for population studies

Rapid Binary Population Synthesis

e.g., COMPAS (Stevenson et al. 2017),
COMBINE (Kruckow et al. 2018)

Caveat : Fail to reproduce the Galactic DNS properties!

e.g., Orbital period – eccentricity distribution



Vigna Gómez et al. (2018)

Project I, Part A

Simulation of Double Neutron Stars

(using detailed stellar evolution code MESA)

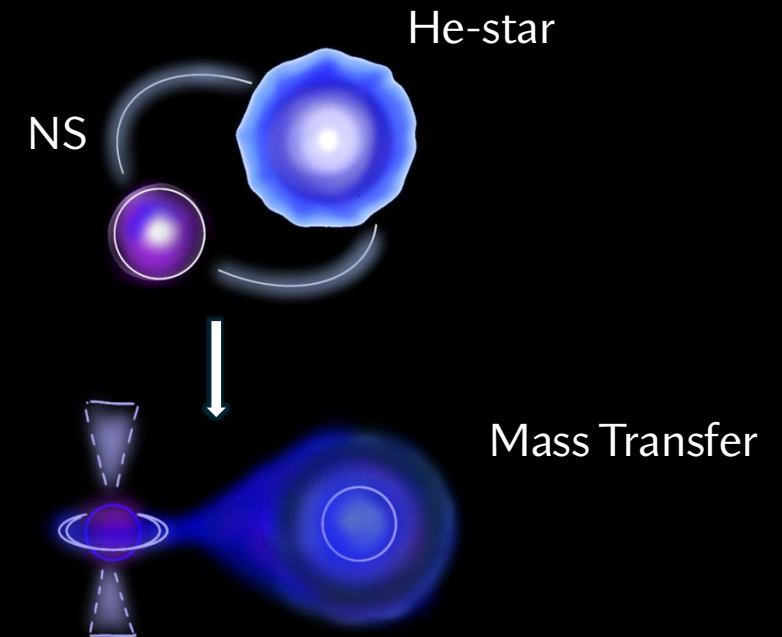
Initial parameter space

He ZAMS mass: $2.5 - 10 M_{\odot}$

First-born NS (NS1) mass: $1.4 M_{\odot}$

Orbital period: 1.5 hrs – 100 days

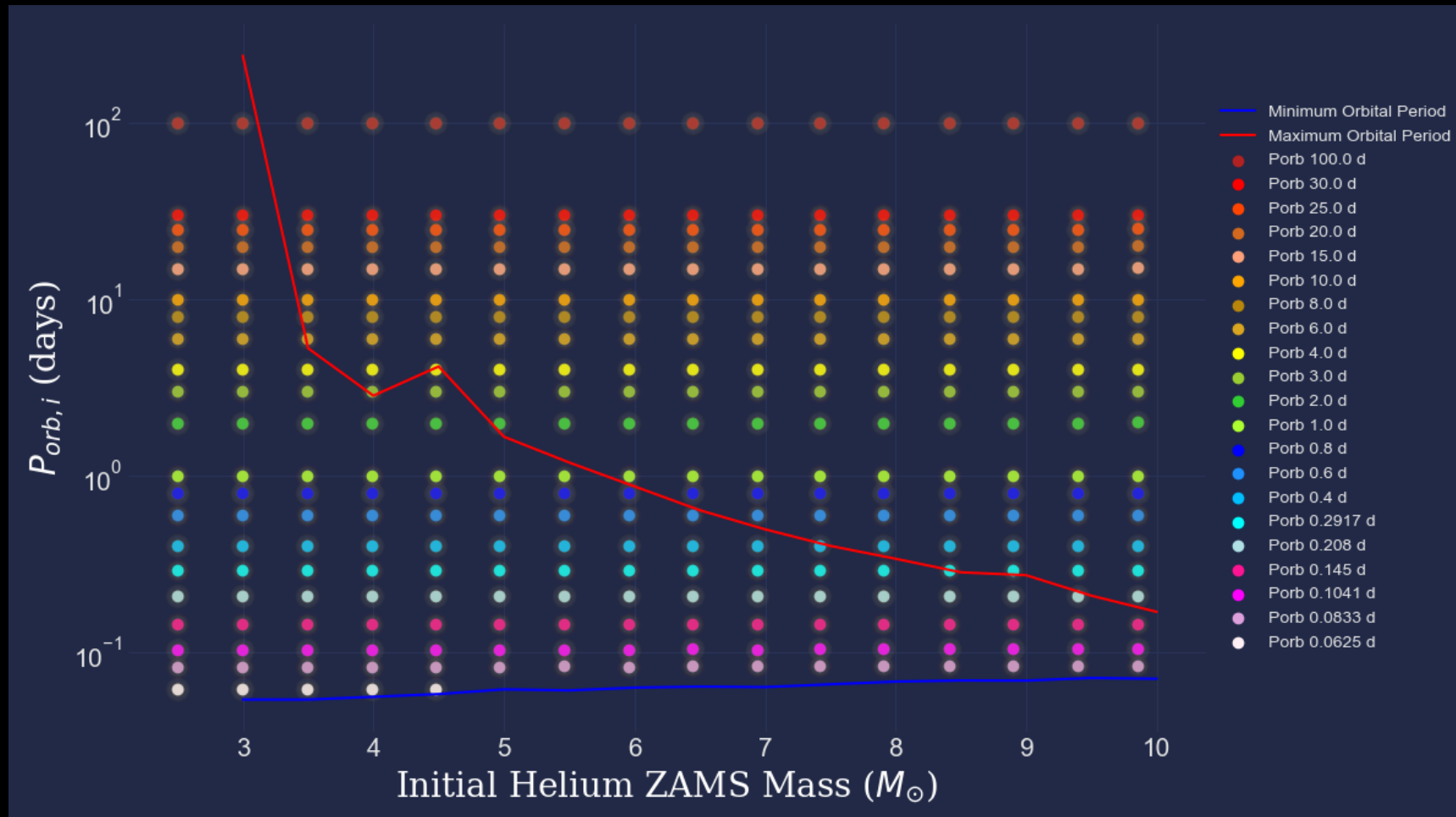
Metallicity: $Z = \text{Solar metallicity } (Z_{\odot})$

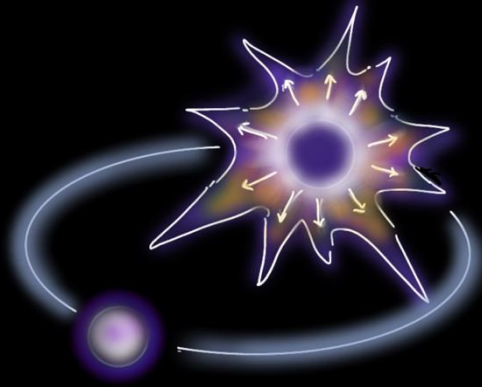


MESA

Paxton et al. 2011

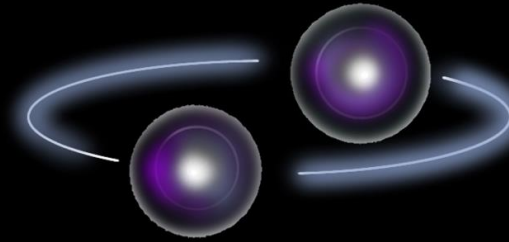
Grid of Models Based On Initial He ZAMS (solar metallicity) Mass + $1.4 M_{\odot}$ NS and Initial Orbital Period



*Project I, Part A***Simulation of Double Neutron Stars**

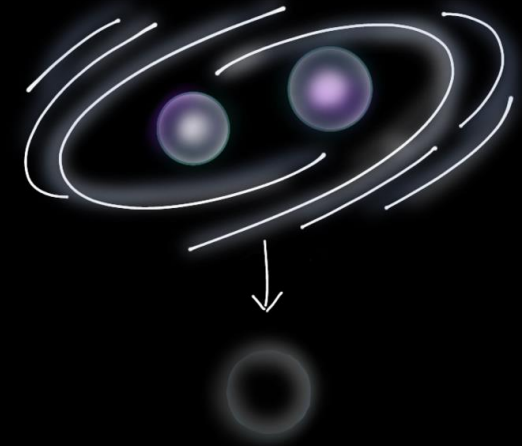
Modelling supernova and kick physics

Mandel & Müller 2020,
Kapil et al. 2023



Post-Supernova effects on orbital dynamics

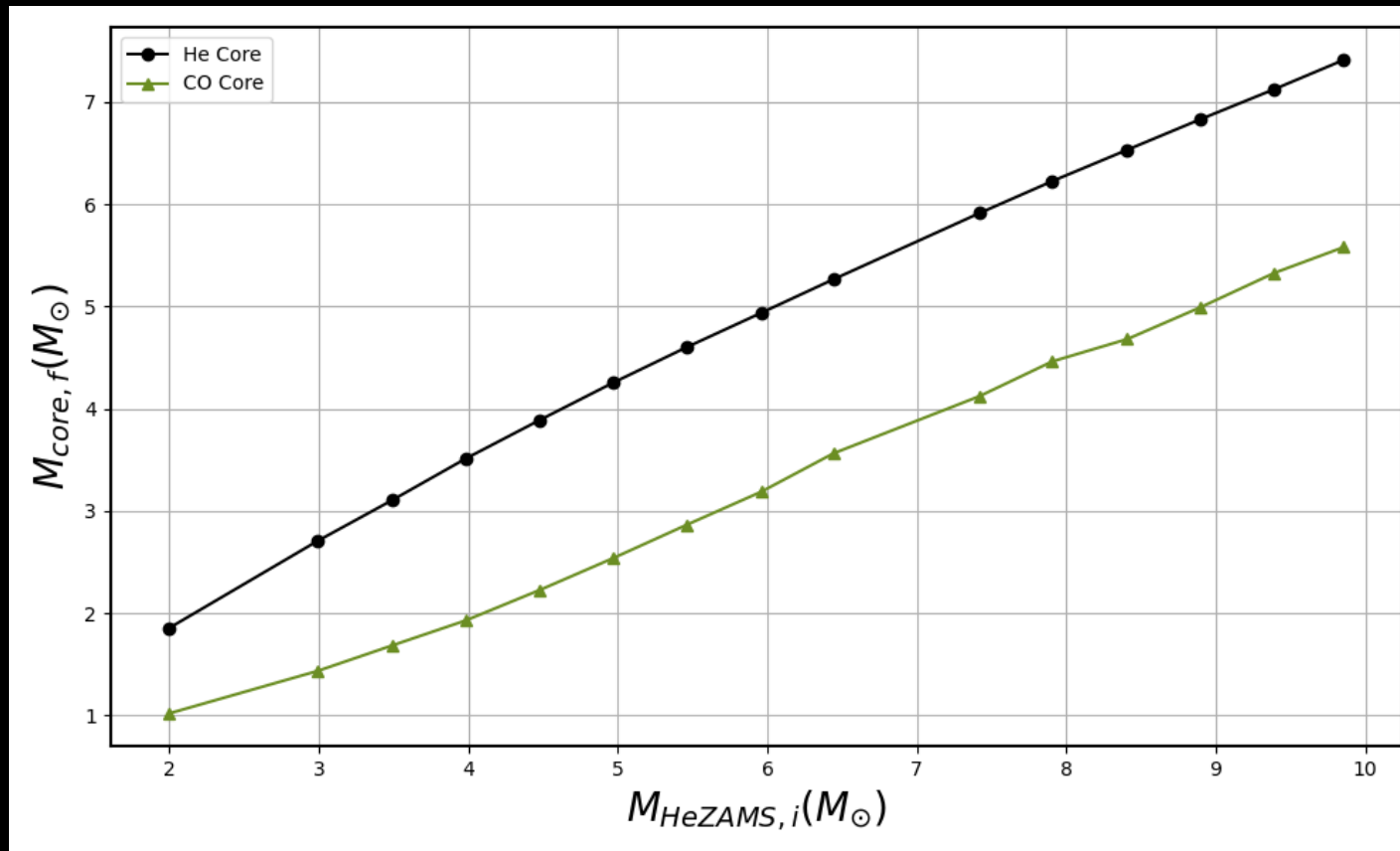
Brandt &
Podsiadlowski 1995



Gravitational-wave merger timescale

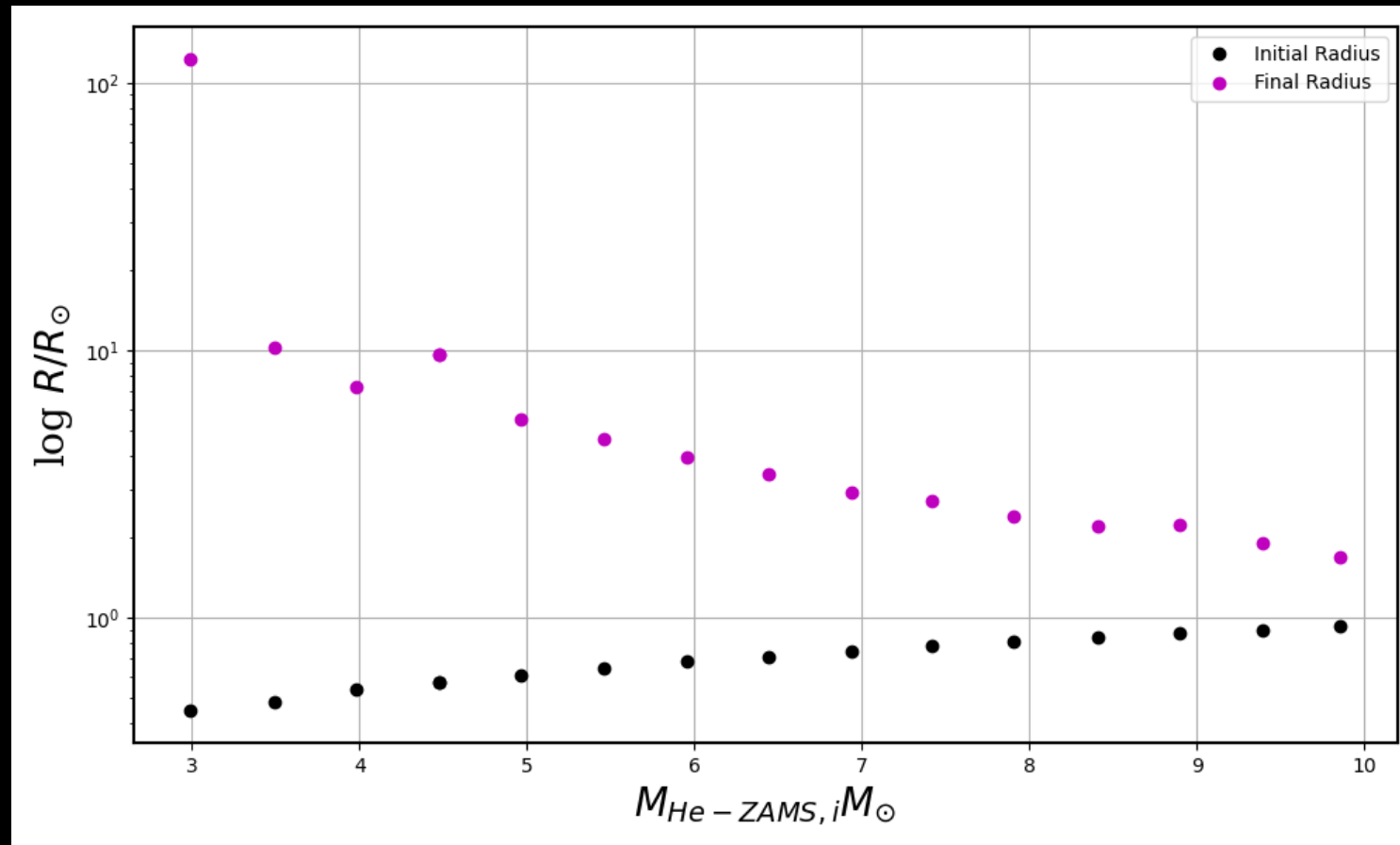
Peters 1964b
Mandel (2021b)

He and carbon-oxygen (CO) core masses of all He-stars

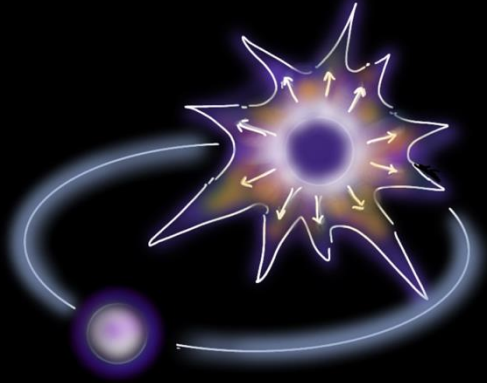


- Core masses are well determined by the star's initial mass and standard choice of stellar physics.

Initial and final radius as a function of initial He-ZAMS mass



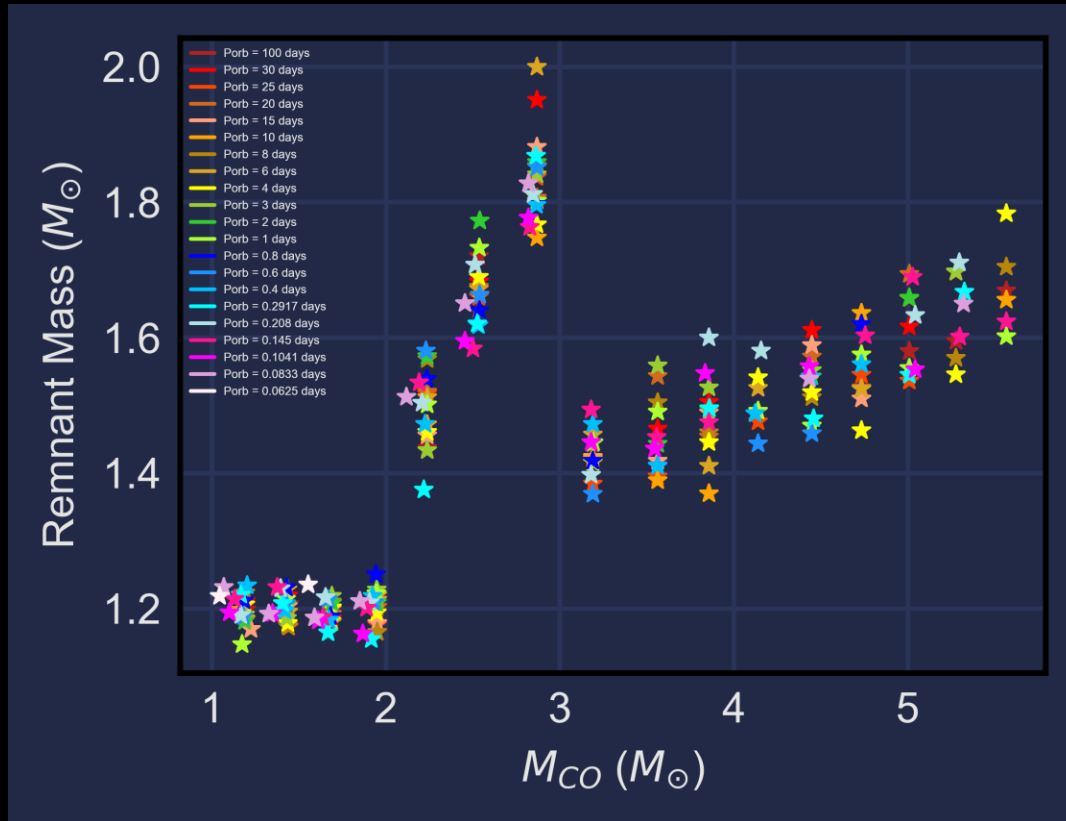
Incorporating supernova and kick prescription



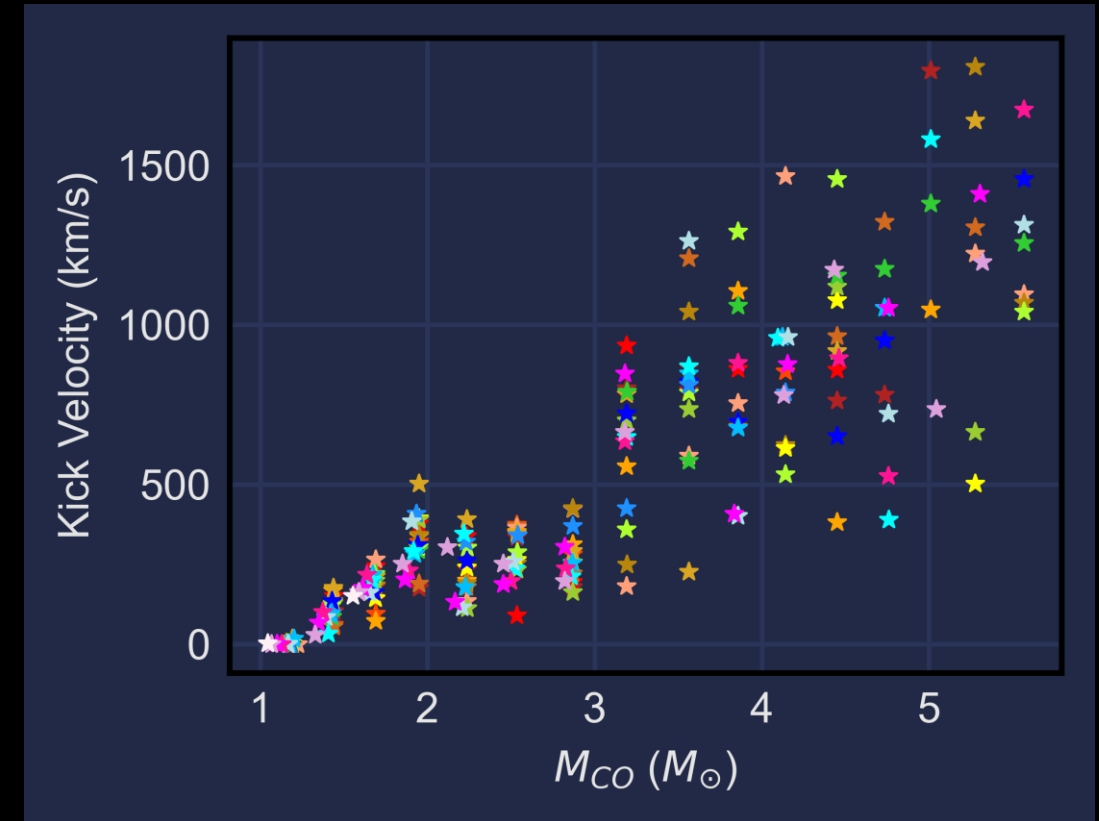
We call this the ‘Standard Model’

- Mandel & Muller (2020) have developed an analytic prescription for the dependence of the mass of NS and BHs and natal kicks based on the pre-SN carbon-oxygen (CO) core and helium shell mass

Standard Supernovae and Kick Prescription (Standard Model)



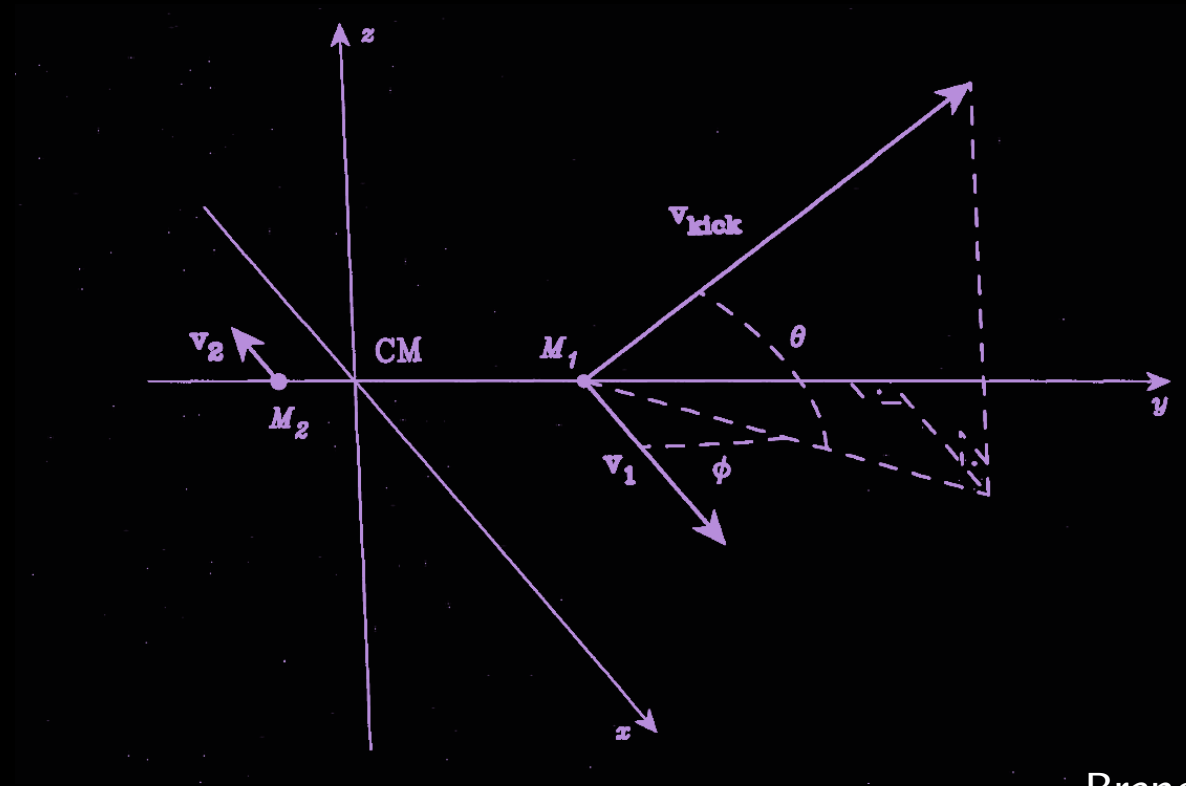
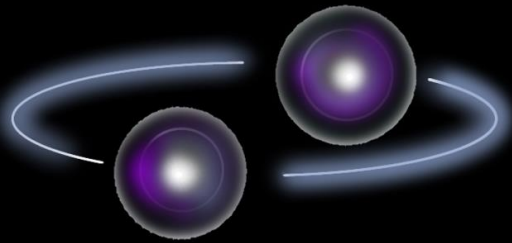
Remnant masses (NS2) as a function of CO core mass



Kick velocity for the corresponding NS2 as a function of CO core mass

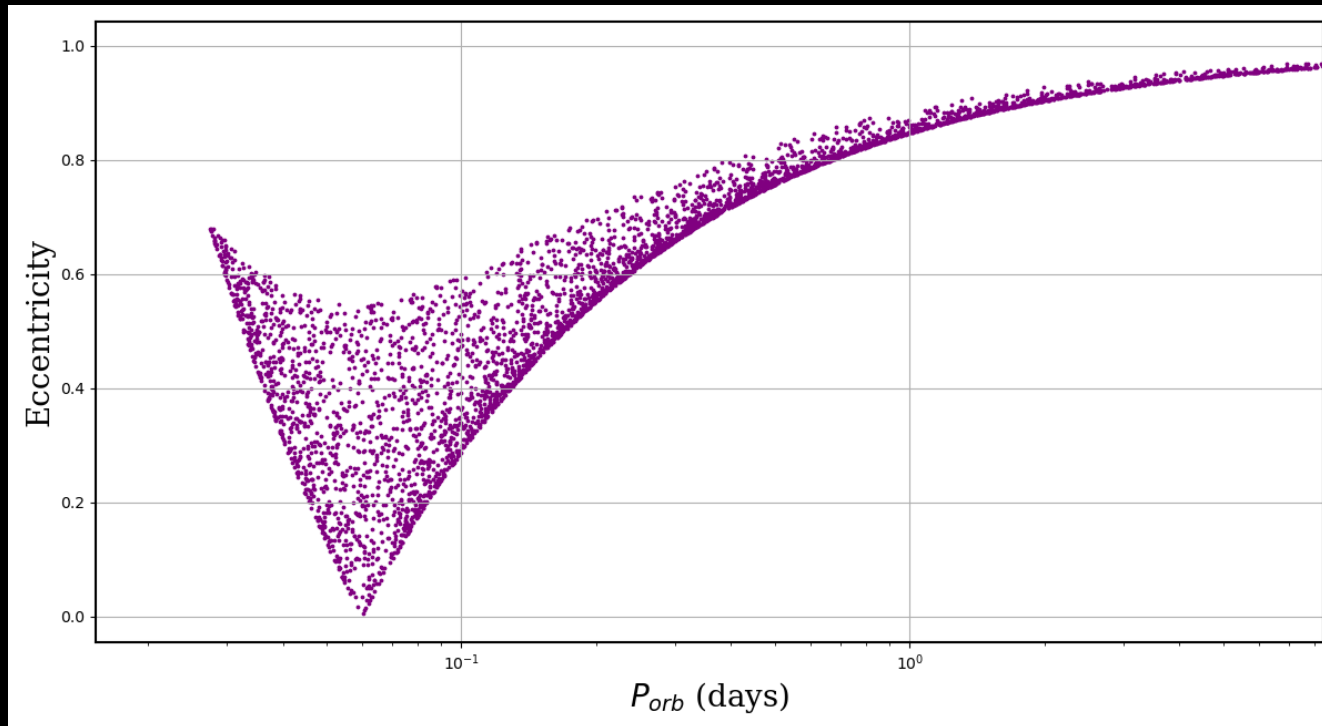
Post-Supernova effects on orbital dynamics

Determine the relation between the initial and final binary orbital parameters and the effect of asymmetric supernovae on binary systems



Brandt et al. 1995

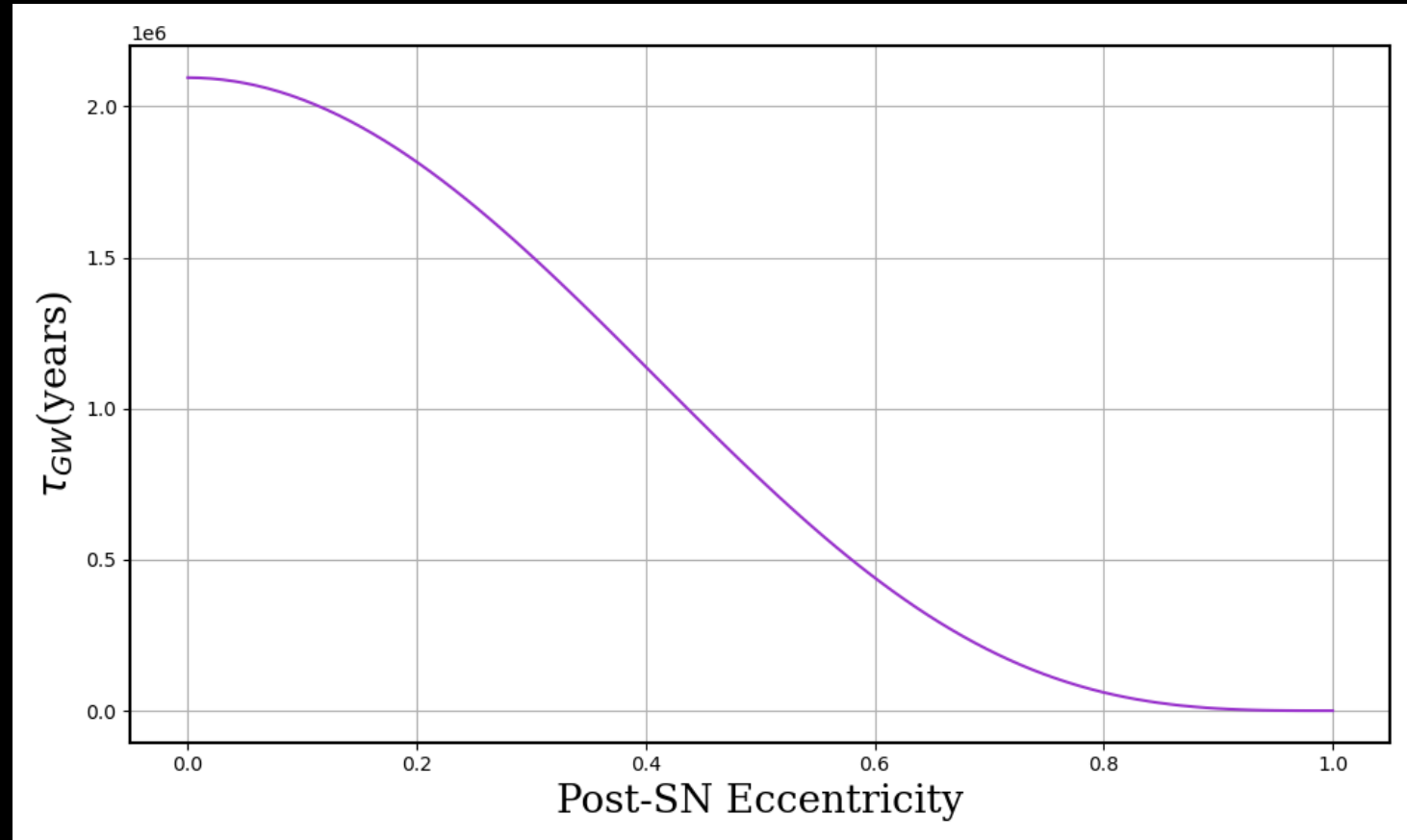
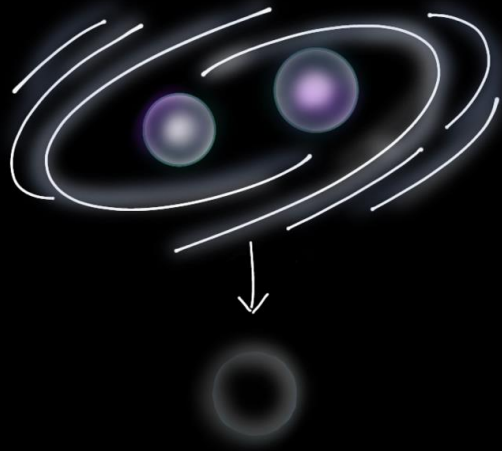
Eccentricity as a function of orbital period for Model 3: $M' = 1.5M_{\odot} + 1.4M_{\odot}$ NS (post-SN binary)



The *post-SN semi-major axis, orbital period*, and the *post-SN eccentricity* determine if the binary will merge within the Hubble time.

Gravitational-wave merger timescale

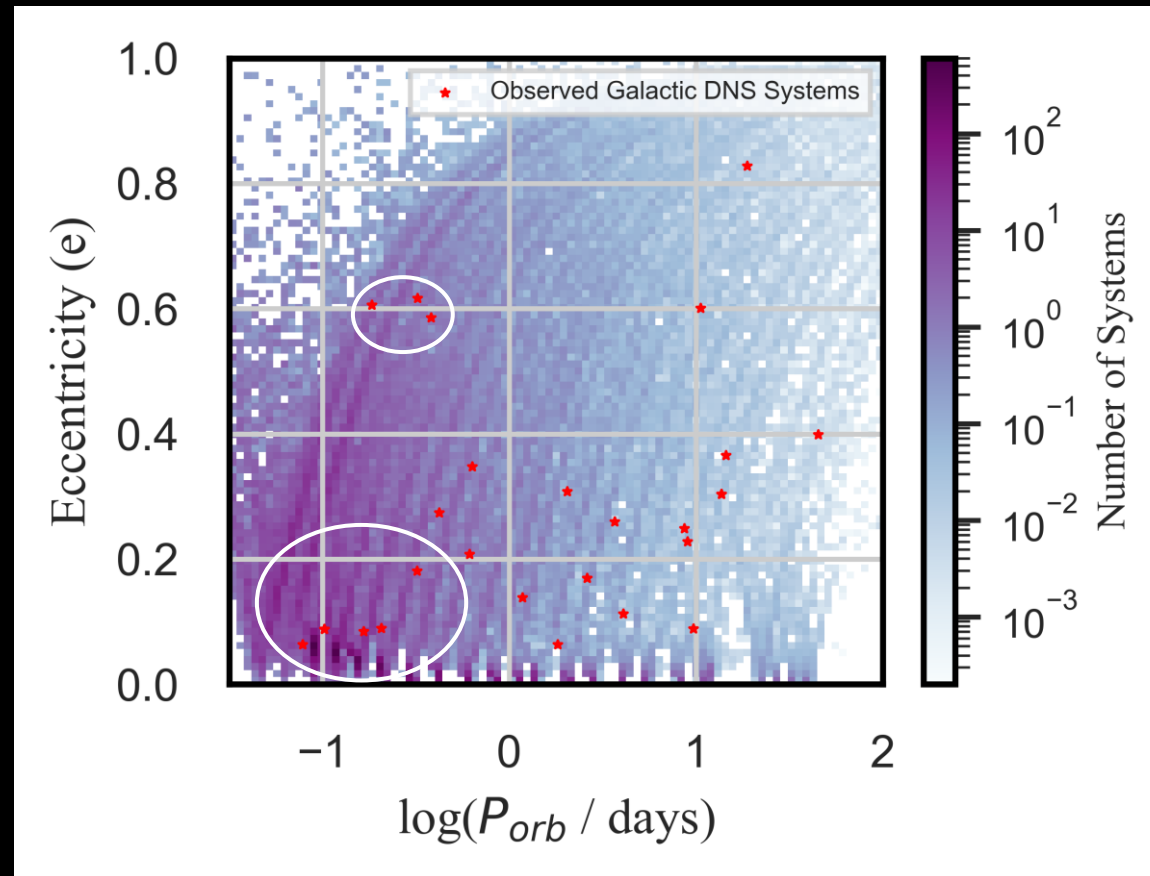
Gravitational wave merger time as a function of post-SN eccentricity for $NS2 = 1.5 M_{\odot} + NS1 = 1.4 M_{\odot}$



Results

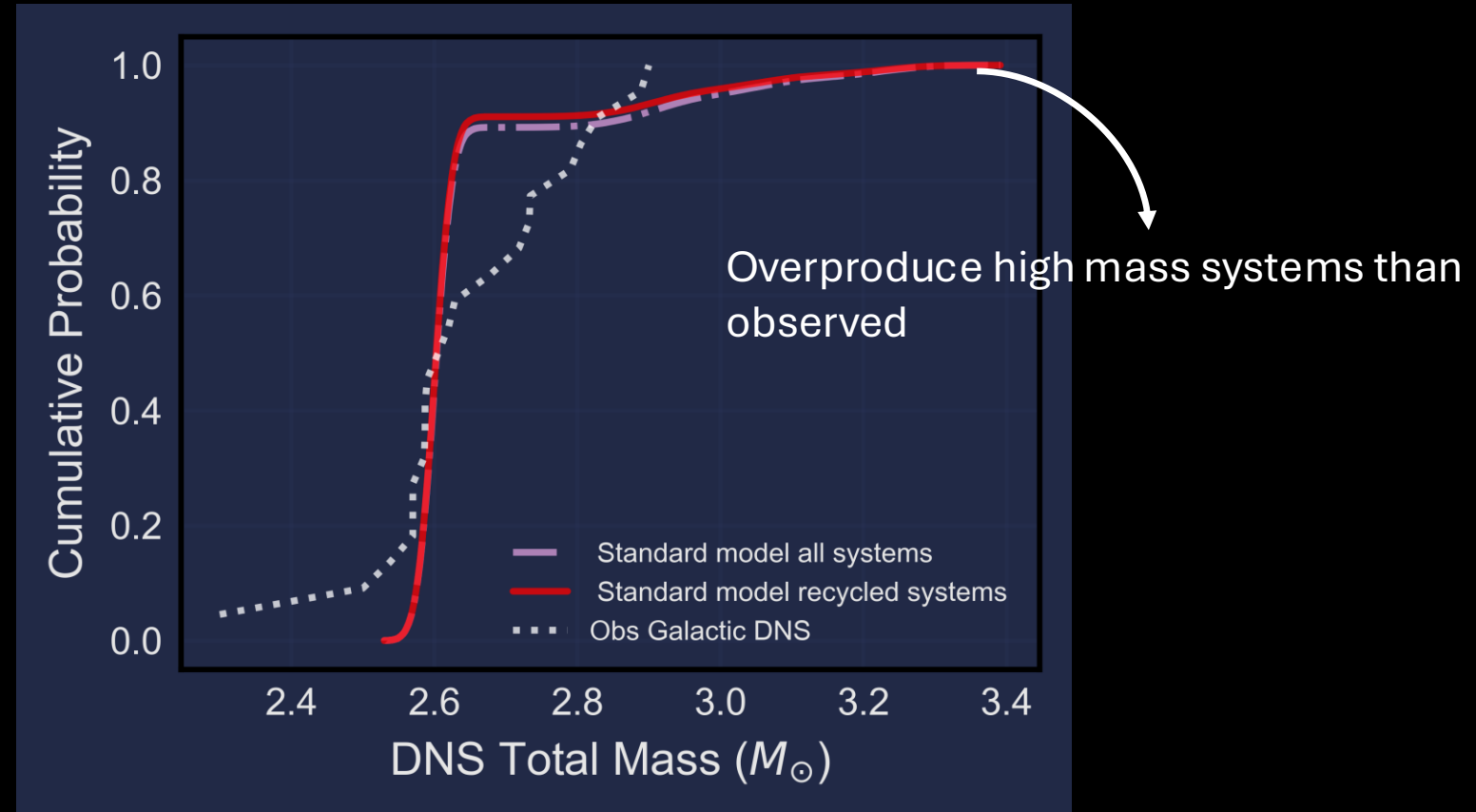
Combined P_{orb} -e Distribution of all models

Standard Model



DNS Total Mass Distribution

Standard Model



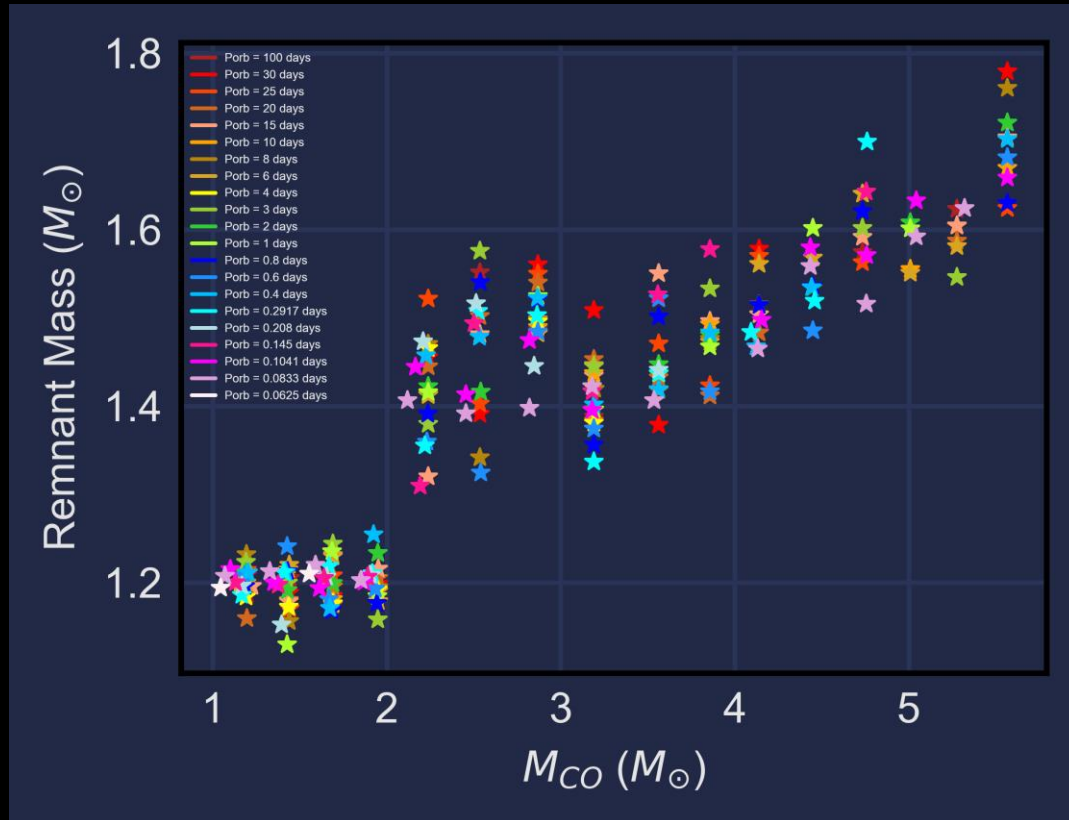
Issues with the Standard Prescription – Mandel & Mueller 2021

- Overpredicts orbital eccentricities of the systems
- Overpredicts DNS total mass compared to what we observe at solar metallicity

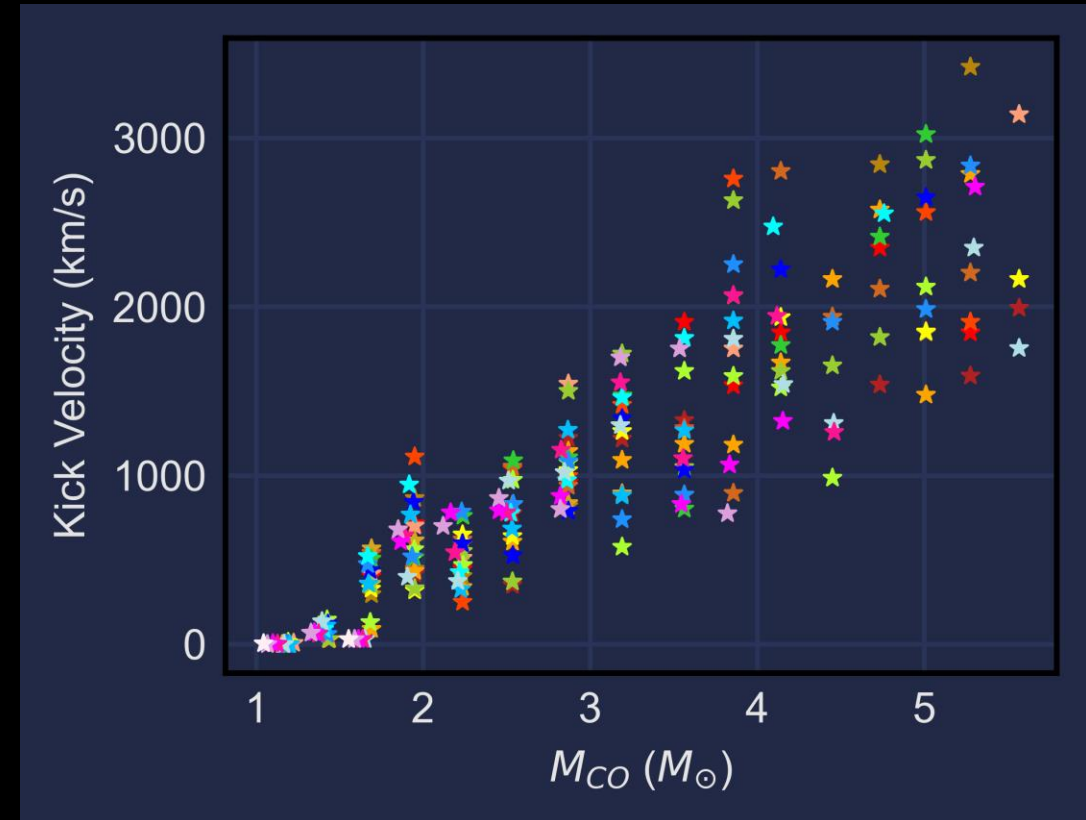
Modifications to the Standard Model

1. Different remnant mass and CO core mass relation
2. Incorporate Ultra-stripped and Electron capture supernova conditions
3. Kick slow merging systems to high velocities

Modified Model



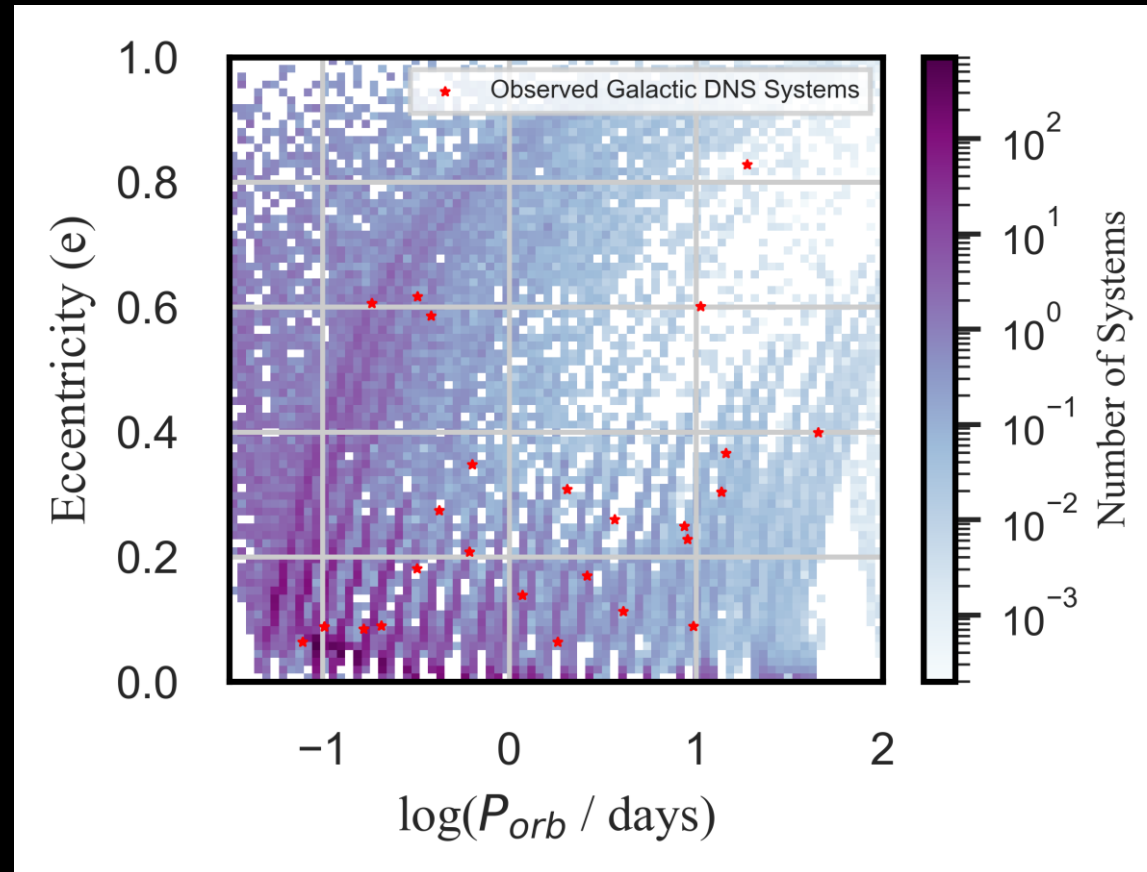
Remnant masses (NS2) as a function of CO core mass



Kick velocity for the corresponding NS2 as a function of CO core mass

Combined Porb-e Distribution of all models

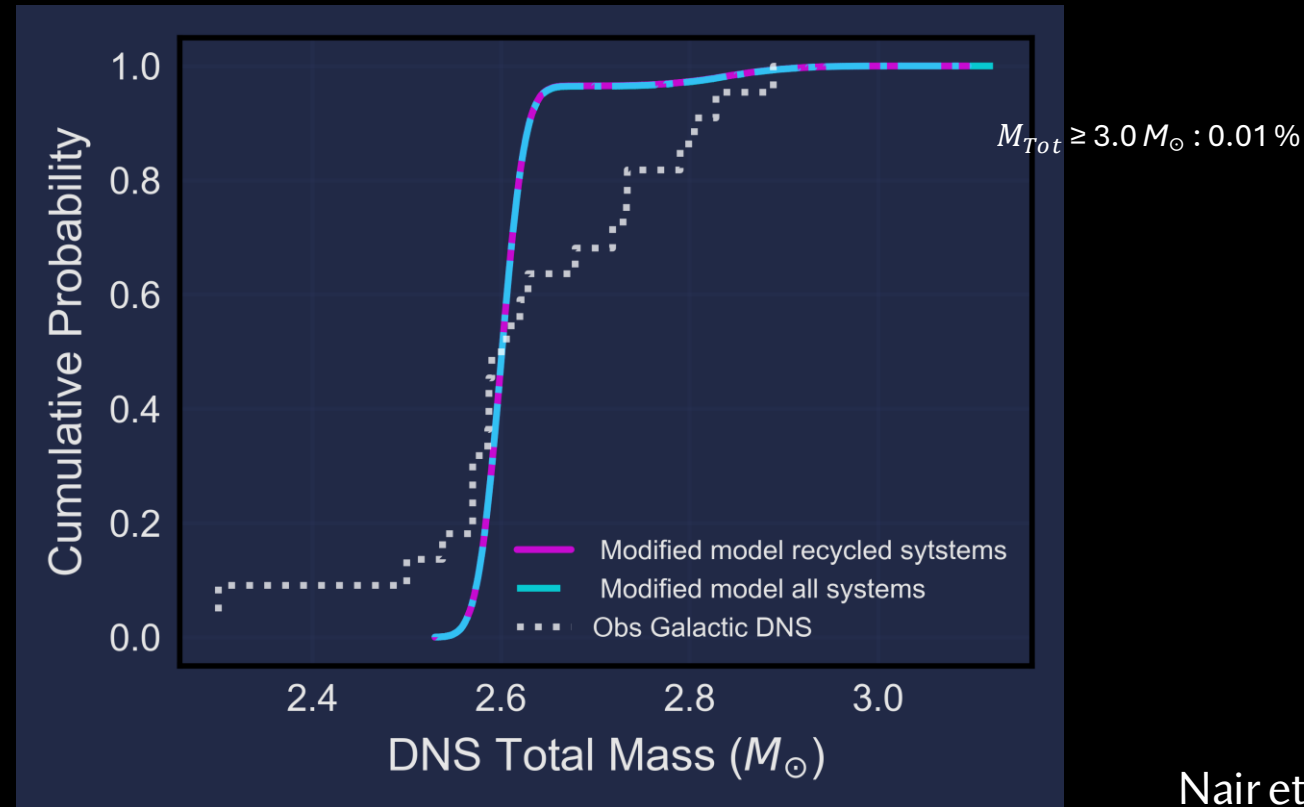
Modified Model



Nair et al. 2025 (Published in MNRAS)

DNS Total Mass Distribution

Modified Model

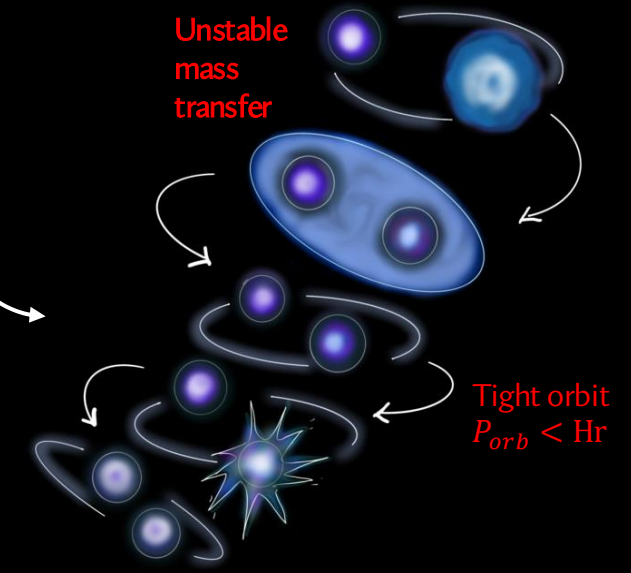
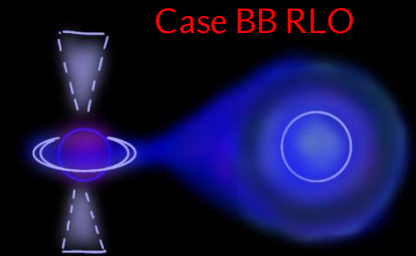
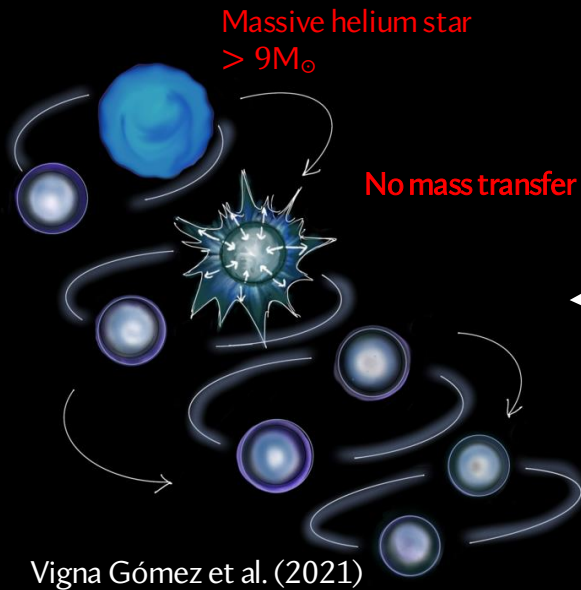


Nair et al. 2025 (Published in MNRAS)

GW190425 Progenitors

Modified: (Out of all the systems $\geq 3 M_{\odot}$)

- 82% of systems - formed via Case BB RLO MT
- 18% of systems - formed via massive helium stars without recycling the first-born NS
- Rules out the formation pathway via unstable mass transfer



Fewer DNS systems $\geq 3 M_{\odot}$

Project I, Part B

Initial parameter space

He ZAMS : $2.5 - 10 M_{\odot}$



First-born NS (NS1) : $1.1 - 1.9 M_{\odot}$

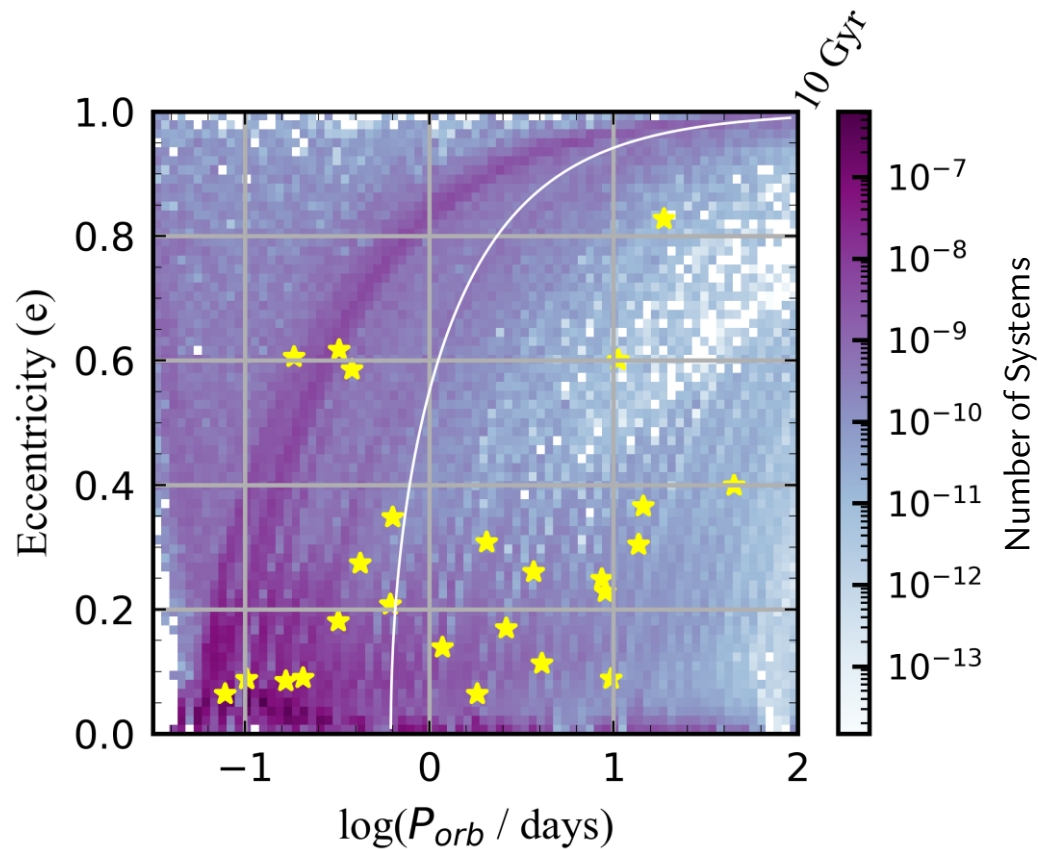
Orbital period : 1.5 hrs – 100 days



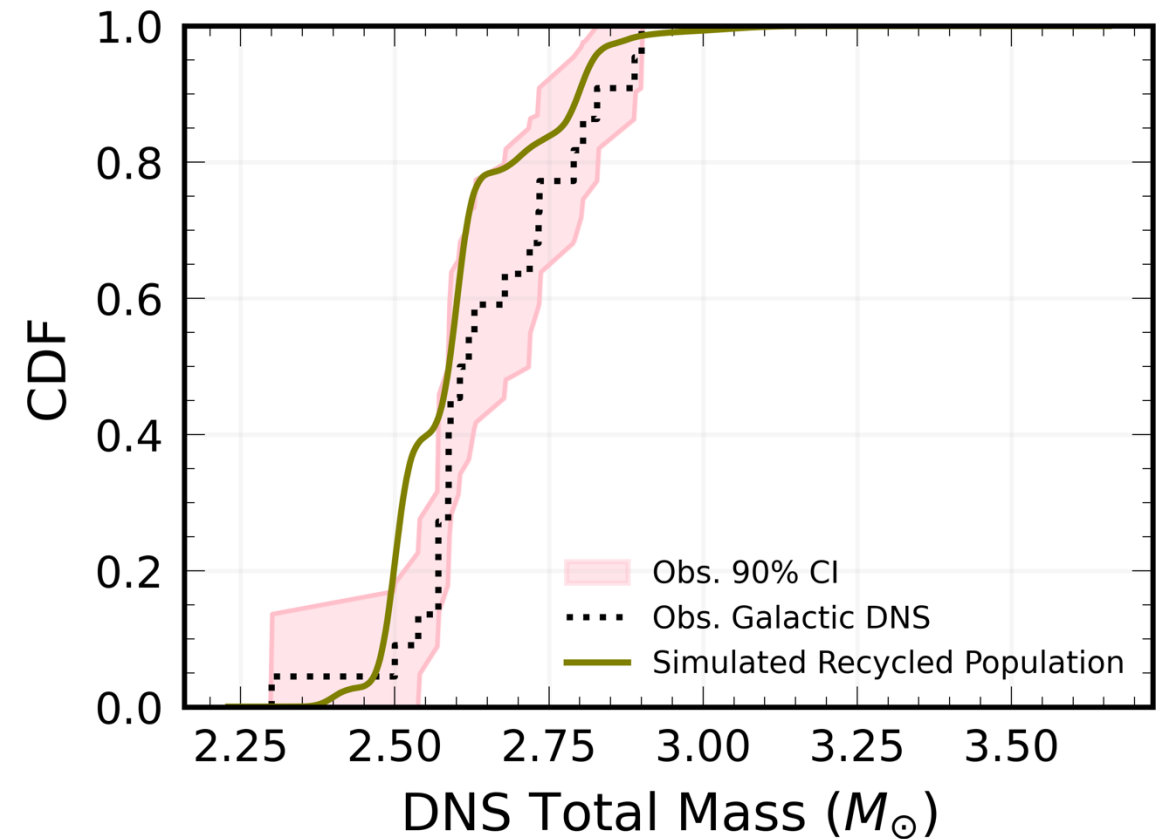
Metallicity : $Z = Z_{\odot}$, $Z = 0.1 Z_{\odot}$ and $Z = 0.01 Z_{\odot}$

Simulated DNS population compared against observed DNS population ($Z = Z_{\odot}$)

Combined Porb-e Distribution of all models



DNS Total Mass Distribution



Formation of Double Neutron Stars at Solar Metallicity

GW190425 Progenitors

a) Standard channel

$\frac{\text{DNS} \geq 3 M_{\odot}}{\text{Total He} - \text{NS binaries}}$

0.47%

We have a winner!

b) Massive He star

0.00006%

Massive He stars and Massive NS1 are rare at solar metallicity ($Z_{\odot}$)!

c) Massive NS1

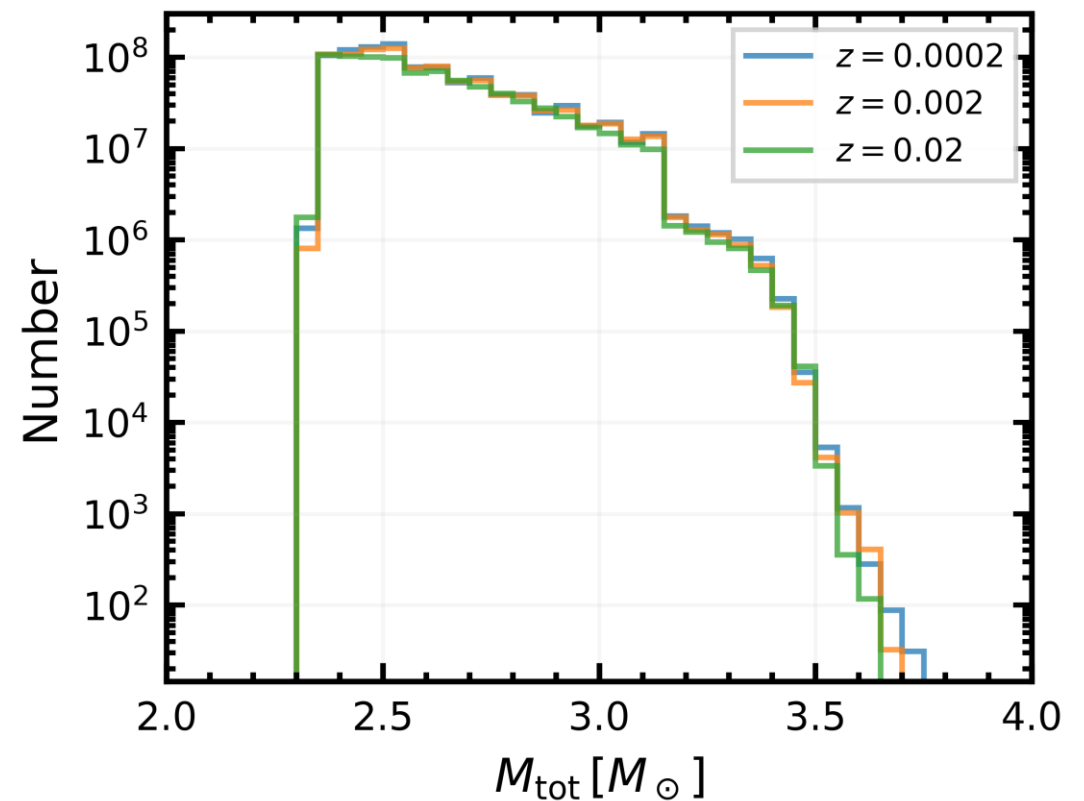
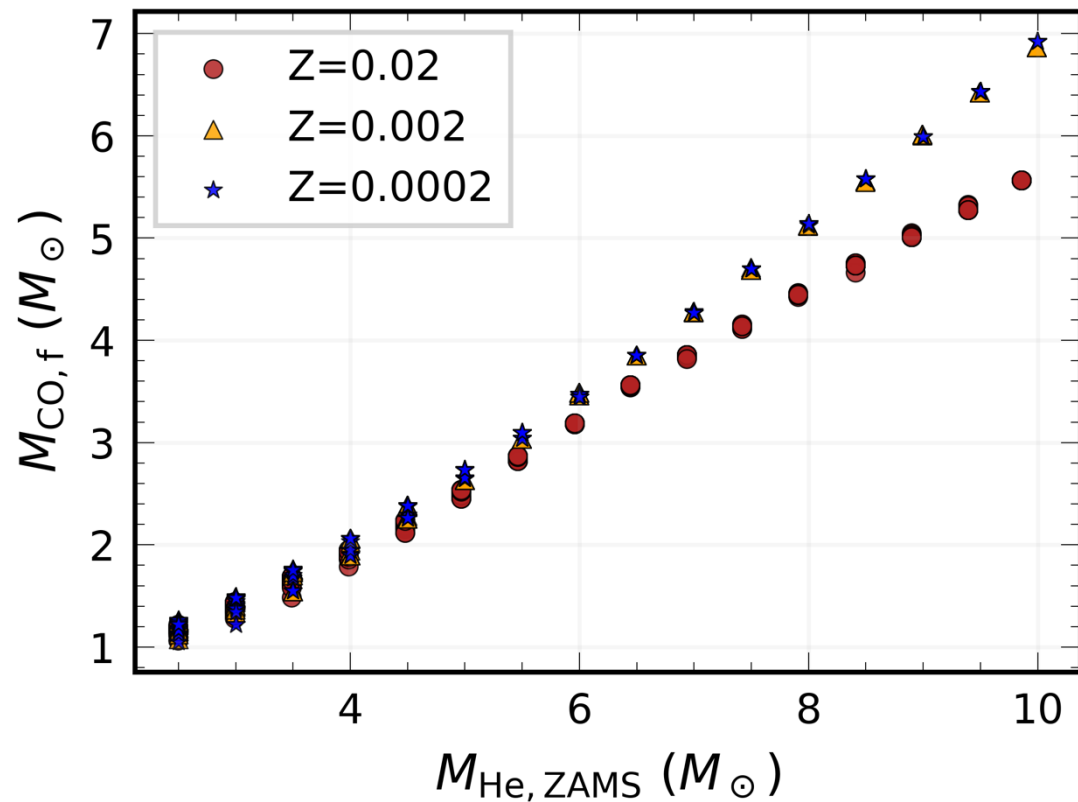
0.0002%

We rule out the “Fast merger” channel at $Z_{\odot}$!

Formation of Double Neutron Stars at Low Metallicity

GW190425 Progenitors

Only 0.5% more than $Z = Z_{\odot}$



MNRAS **000**, 1–8 (2026)

Preprint 9 May 2026

Compiled using MNRAS L^AT_EX style file v3.3

Formation of heavy double neutron stars II: the role of heavy first-born neutron stars and low metallicity

Ashwathi Nair^{1,2★}, Simon Stevenson^{1,2}, Author list TBD^{2,3}¹*OzGrav : The ARC Centre of Excellence for Gravitational Wave Discovery, Hawthorn, VIC 3122, Australia*²*Centre for Astrophysics and Supercomputing, Swinburne University of Technology, Hawthorn, VIC 3122, Australia*

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

In the first paper of this series, we studied the evolution of a $1.4\ M_{\odot}$ neutron star (NS) in a binary with a $2.5\text{--}10\ M_{\odot}$ helium star at solar metallicity ($Z = Z_{\odot}$). In this second paper, we explore the evolution of NS-He star binaries with a broad range of NS masses from $1.1\text{--}1.9\ M_{\odot}$ at $Z = Z_{\odot}$, $Z = 0.1Z_{\odot}$ and $Z = 0.01Z_{\odot}$.

Key words: gravitational waves – stars: massive – stars: evolution – binary evolution

Nair et al. 2026 (in prep)

Work done so far, summarized

- Calibrated models to explain the Galactic DNS population
- Expanded the initial parameter space
- Tested the formation channels to identify dominant GW190425 progenitors
- Standard formation pathway dominates the formation of GW190425 like heavy systems at $Z = Z_{\odot}$
- Rules out formation through unstable mass transfer at solar metallicity
- Massive DNS formation weakly affected by metallicity

Takeaways

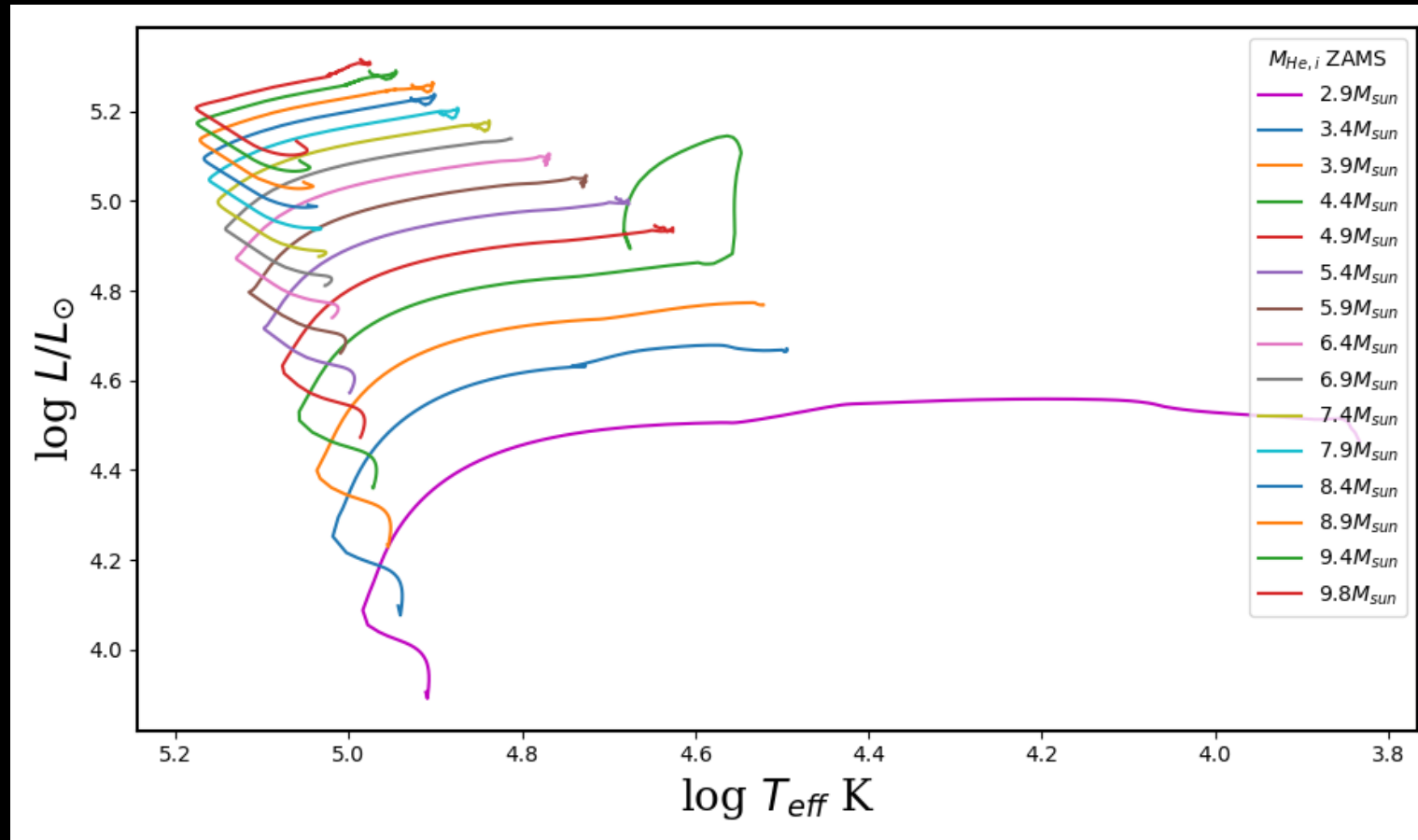
Maybe we don't need a new formation channel to explain the formation of GW190425!

Weak dependence of metallicity on the formation of heavy systems (only 0.5% more systems)

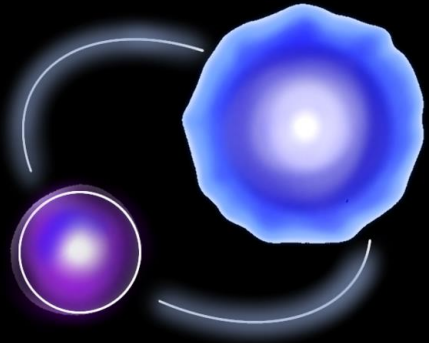
Need more observations !

EXTRA SLIDES

Hertzsprung-Russell (HR) diagram of single helium stars with masses ranging from $2.9M_{\odot}$ - $9.8M_{\odot}$

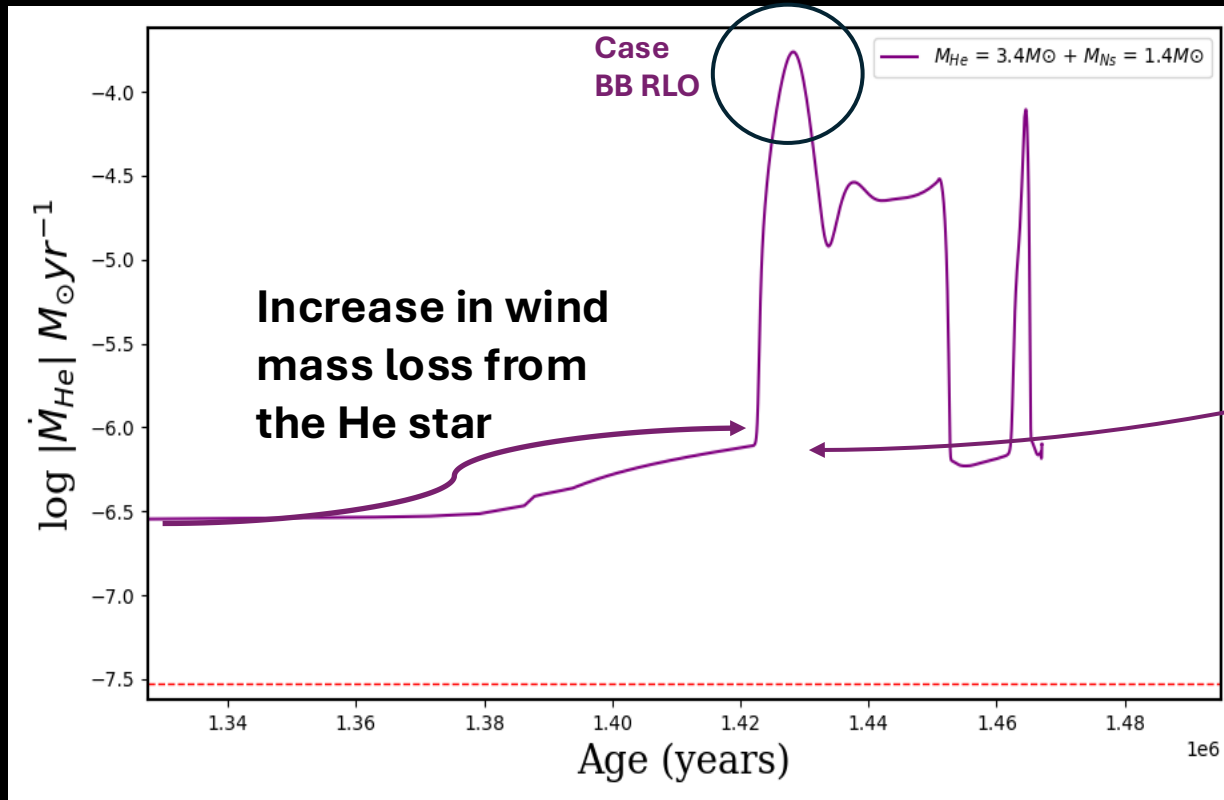


Binary Star Evolution

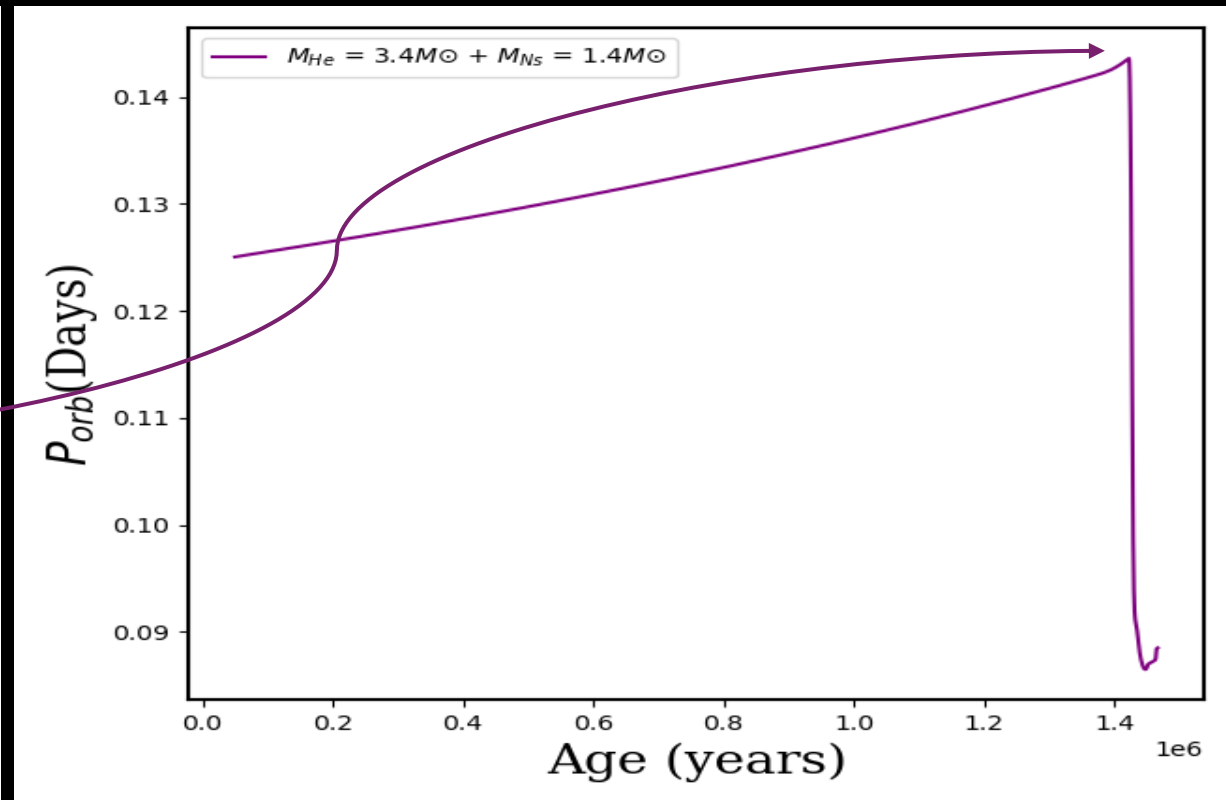


- Modelled the binary evolution of a ZAMS He-star with a $1.4M_{\odot}$ NS companion using MESAbinary
- The binary is initially set up in a circular orbit, with an initial orbital period of 3 hours.
- 'Ritter' mass transfer scheme (Ritter 1988) is used to calculate the mass transfer rate from the He star.

Wind mass loss from the He star as a function of the star's age



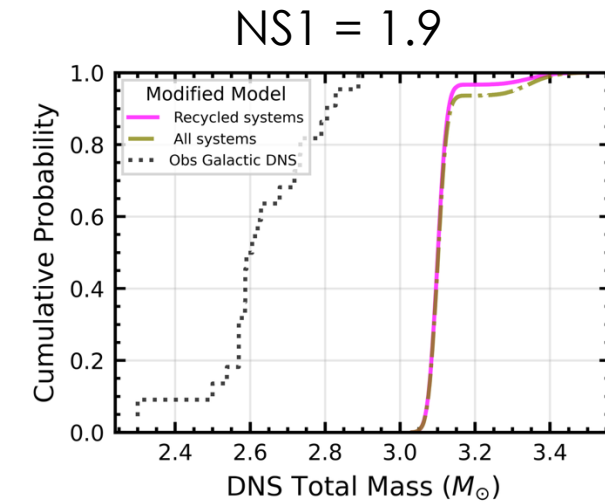
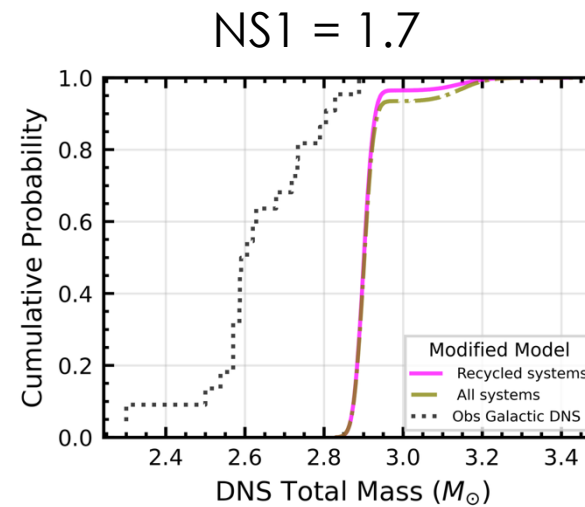
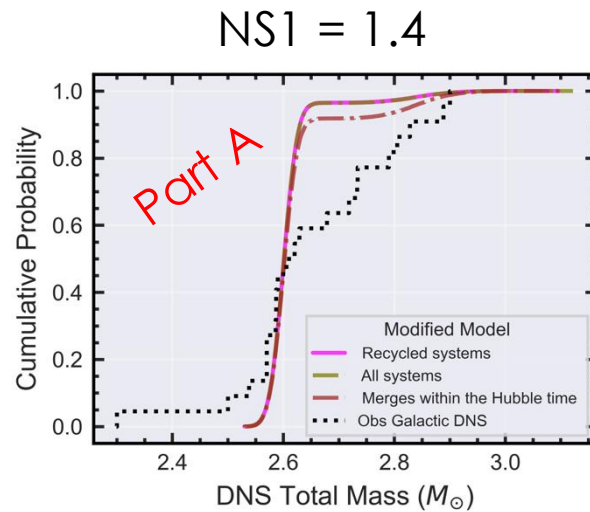
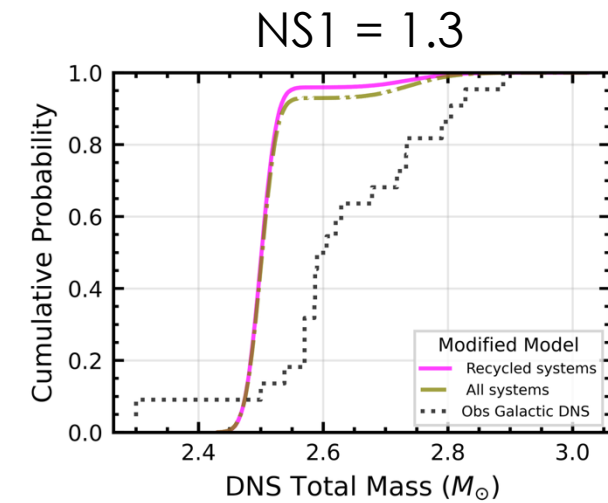
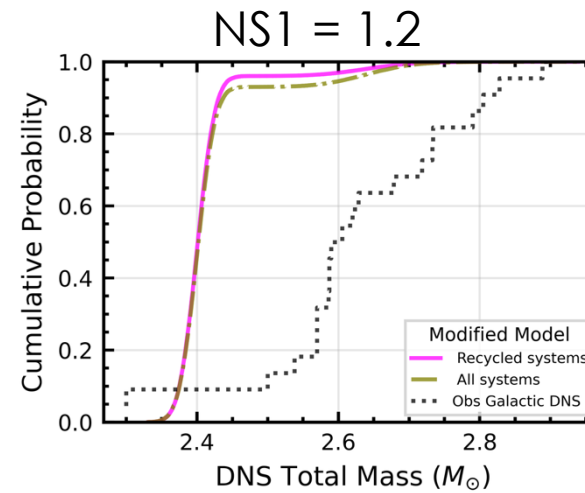
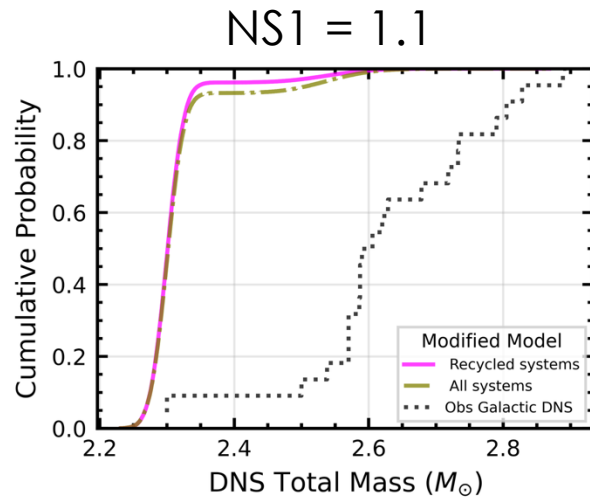
Change in orbital period as a function of the star's age



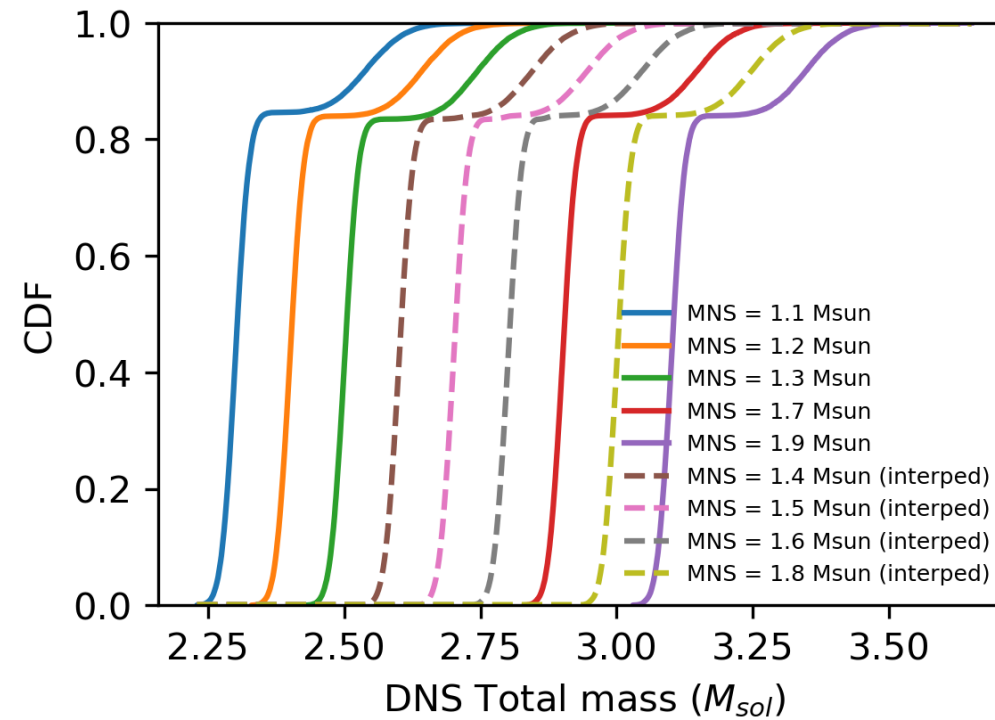
Project 1, Part B

DNS Total mass distribution

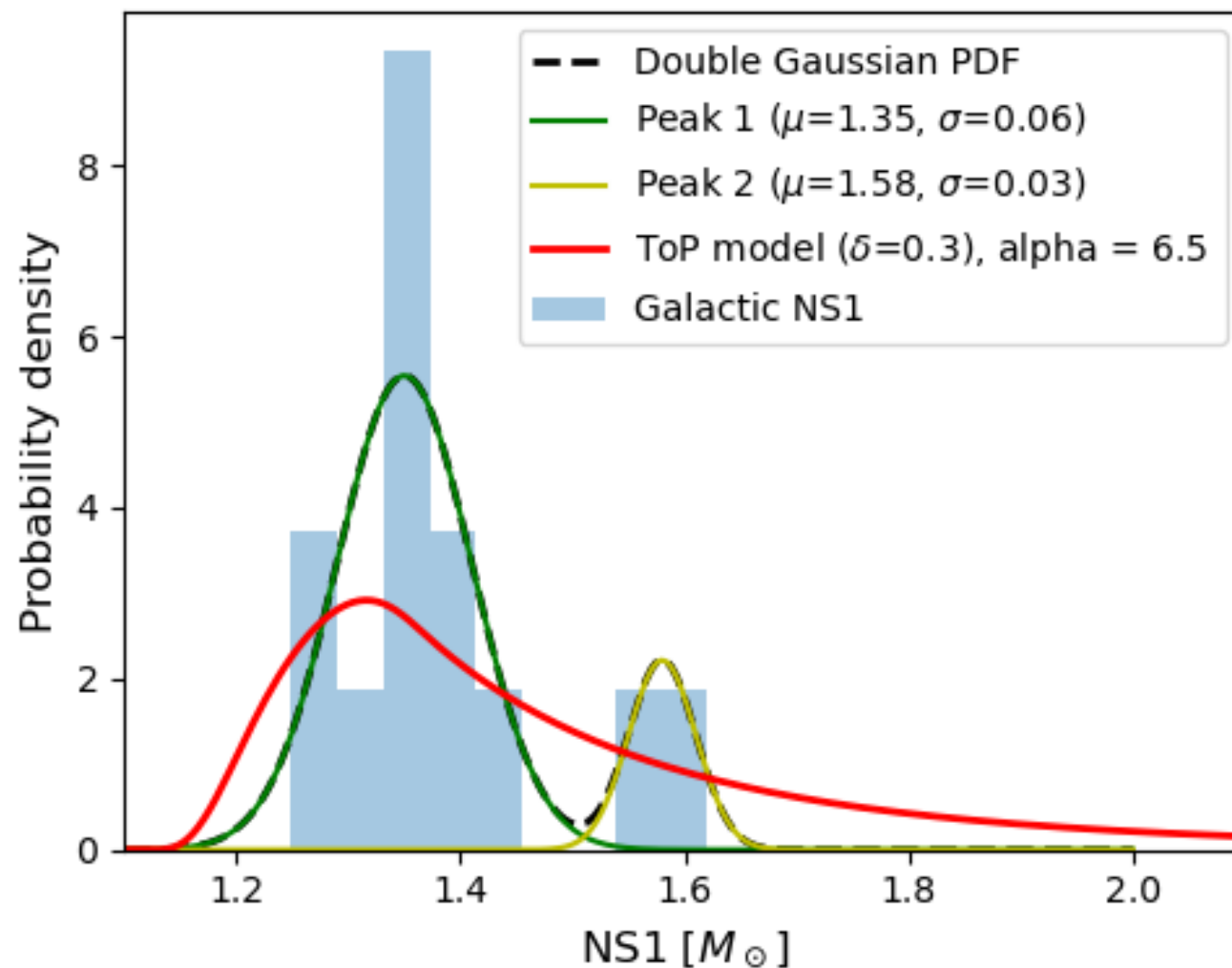
Constant NS1 (1.1, 1.2, 1.3, 1.4, 1.7, 1.9, respectively) + NS2



Interpolated CDFs based on the simulated ones

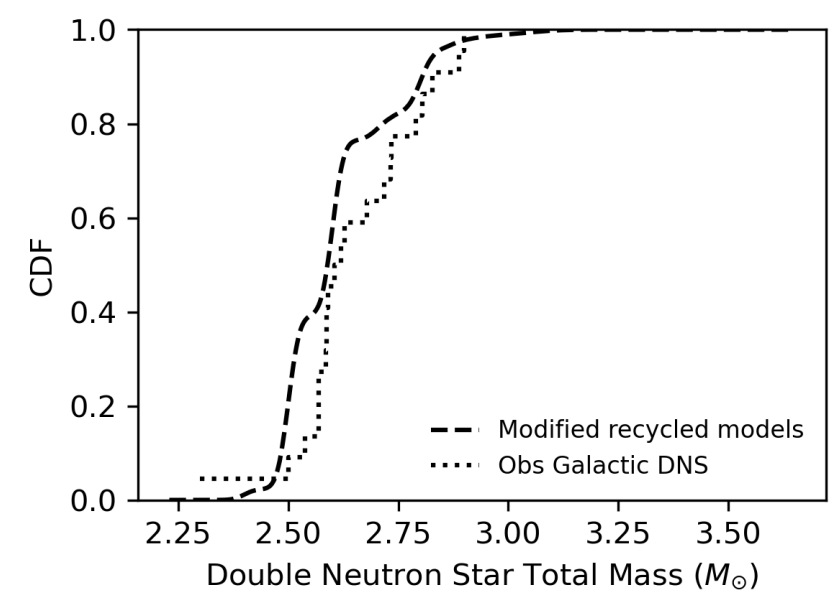


Weights from observed Galactic recycled NSs (double gaussian) and birth-mass function of neutron stars based on the turn-on power law (TOP) model from Z. You et. al 2025

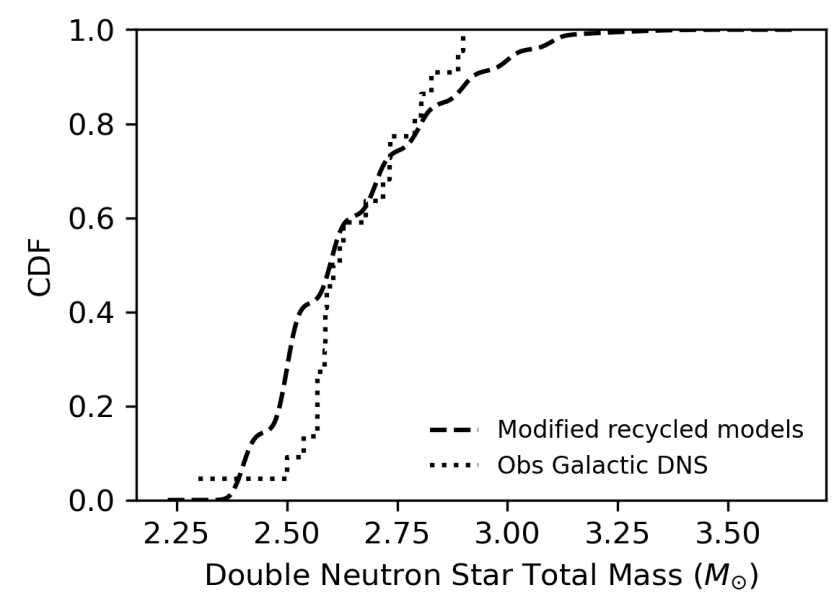


Based on weights from

Observed Galactic DNS

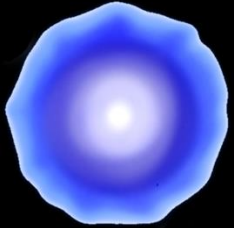


TOP Model
Z. You et al. 2025



Recycled models (simulated) = representative of radio observed population.

Single Helium Star Evolution

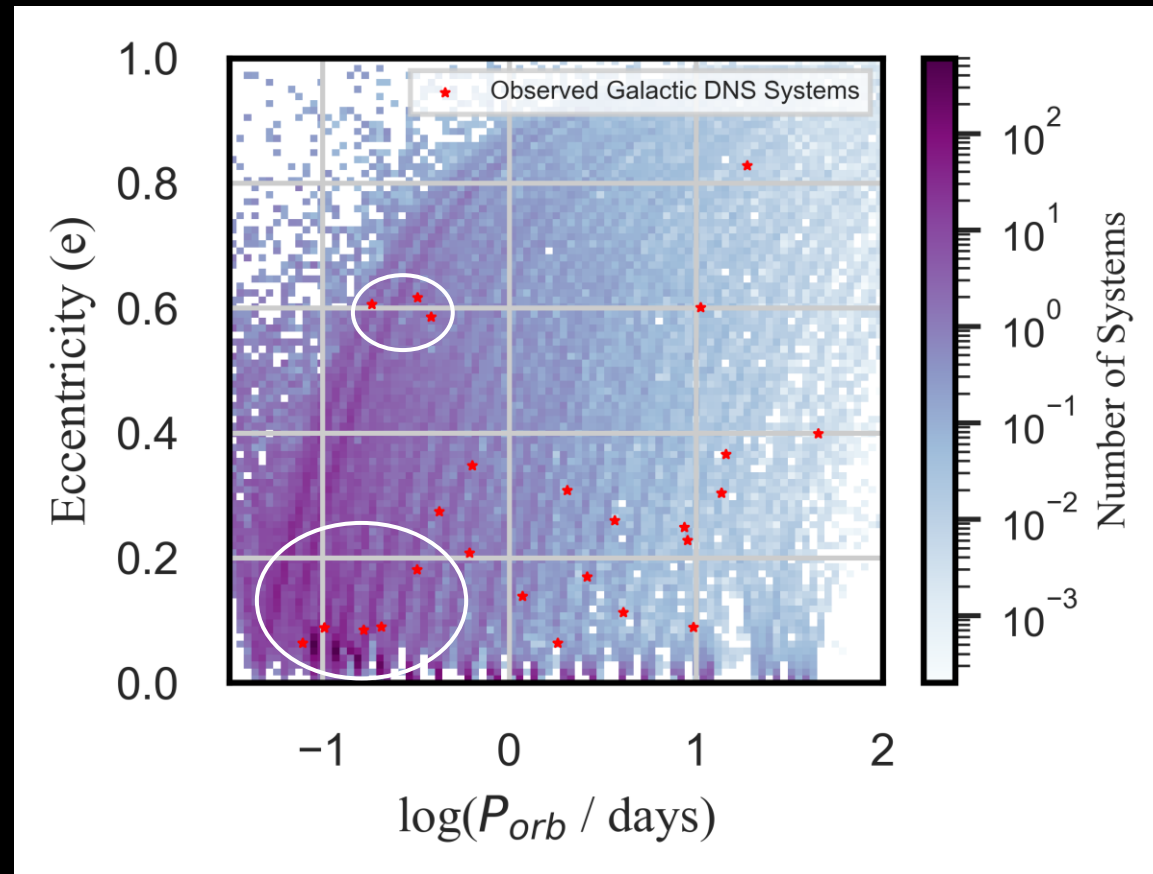


- **Simulations were performed using Module for Experiments in Stellar Astrophysics (MESA) module MESARstar for intermediate and massive He-star ($2.9\text{-}9.8 M_{\odot}$)**
- Helium mass fraction $Y = 0.98$, at solar metallicity $Z = 0.02$
- 'Dutch' prescription for mass loss by hot winds (Nugis & Lamers 2000; Vink et al. 2001; Glebbeek et al. 2009)- scaling factor of 0.8 for massive helium stars (Meynet & Maeder 2000).
- Ledoux criterion for convection, with a mixing length alpha of factor 2, and alpha semi-convection of 1.
- Gold tolerances for intermediate and massive Helium star models, Type2 opacities for $Z = 0.02$

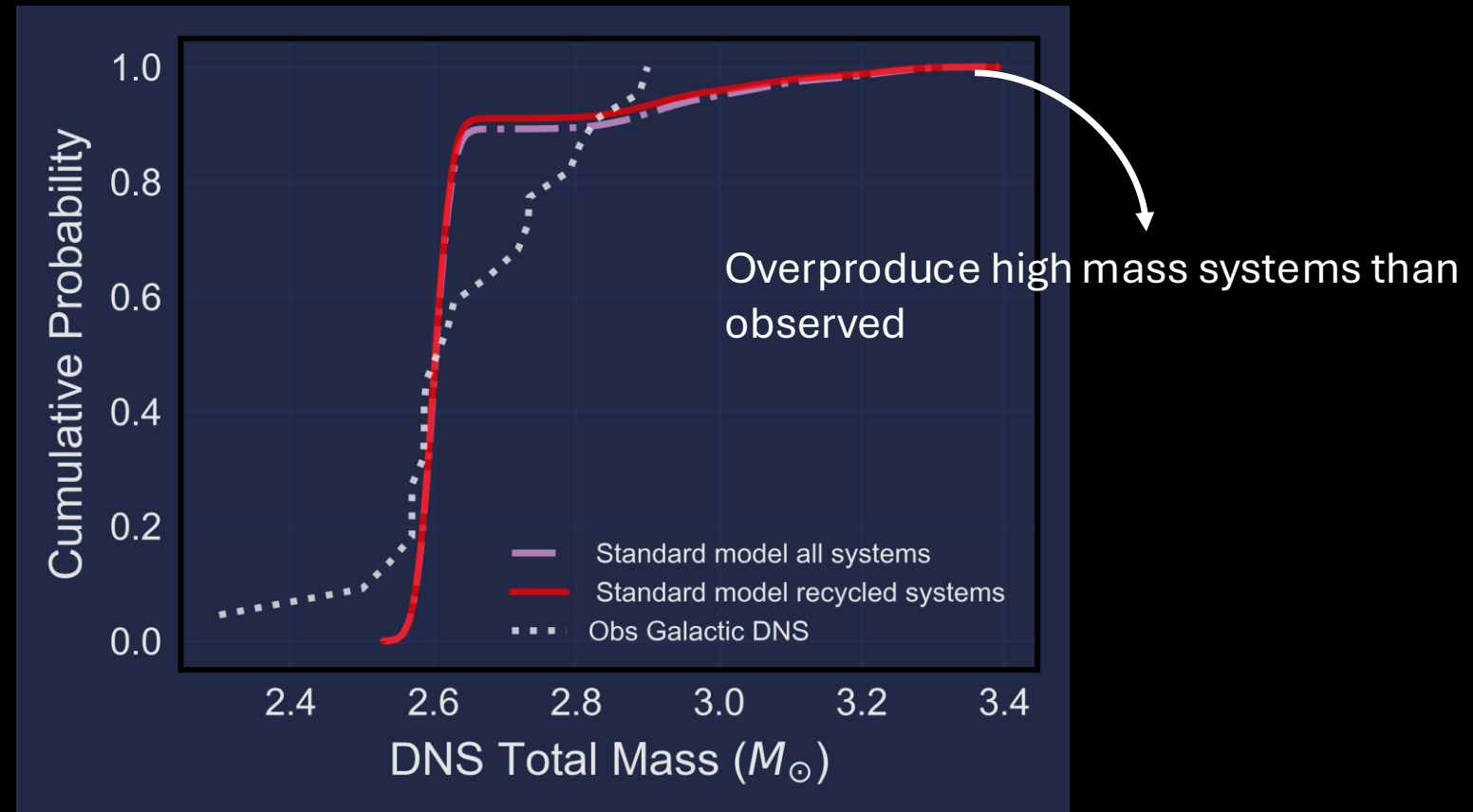


Combined P_{orb} -e Distribution of all models

Standard Model



DNS Total Mass Distribution Standard Model



Issues with the Standard Prescription:

- Overpredicts eccentricities
- Overpredicts DNS total mass within the recycled systems

Modifications to the Standard Model

Why?

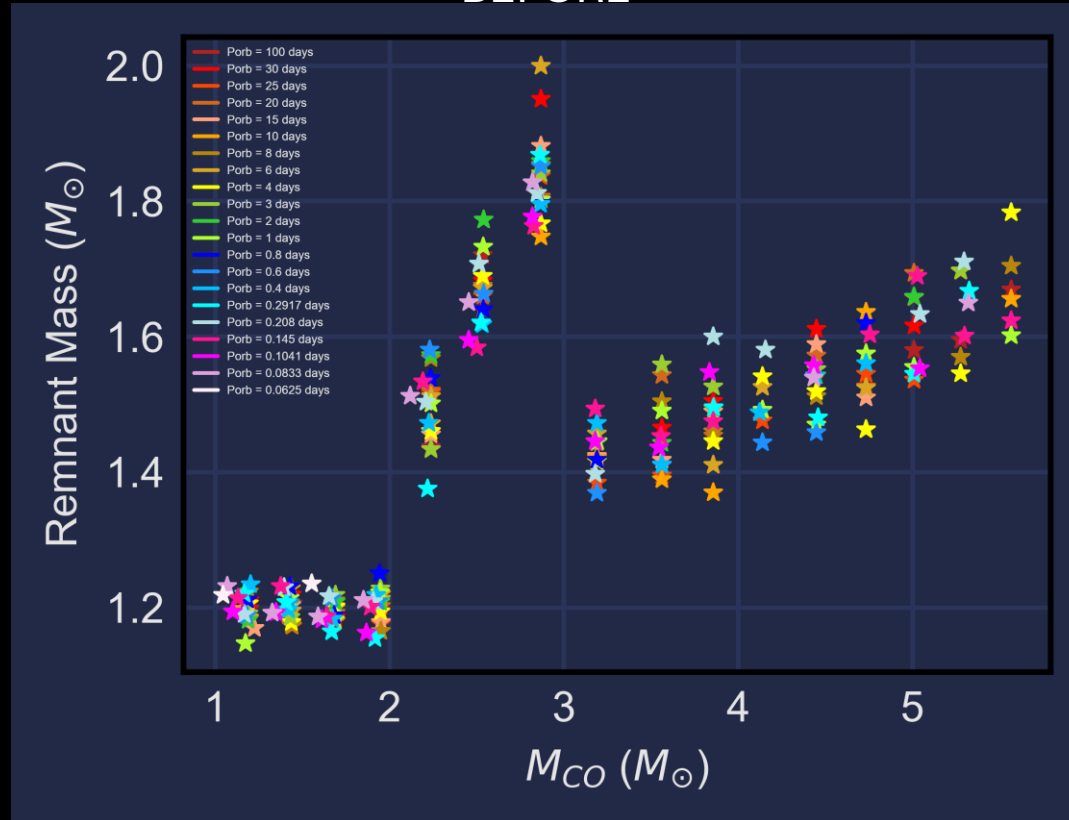
To calibrate my models to explain the observed galactic DNS population

What?

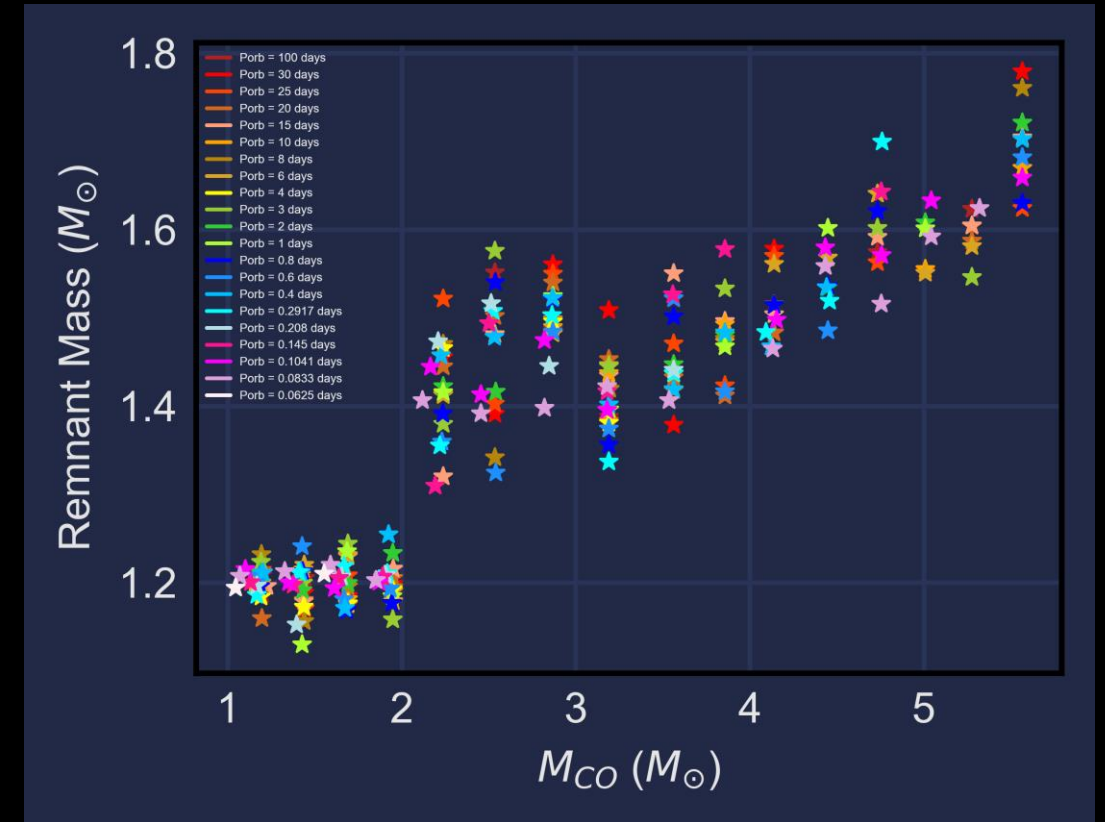
1. The NS scaling factor
2. Defined USSN and ECSN
3. High velocity kicks to slow merging systems

Standard Model

BEFORE



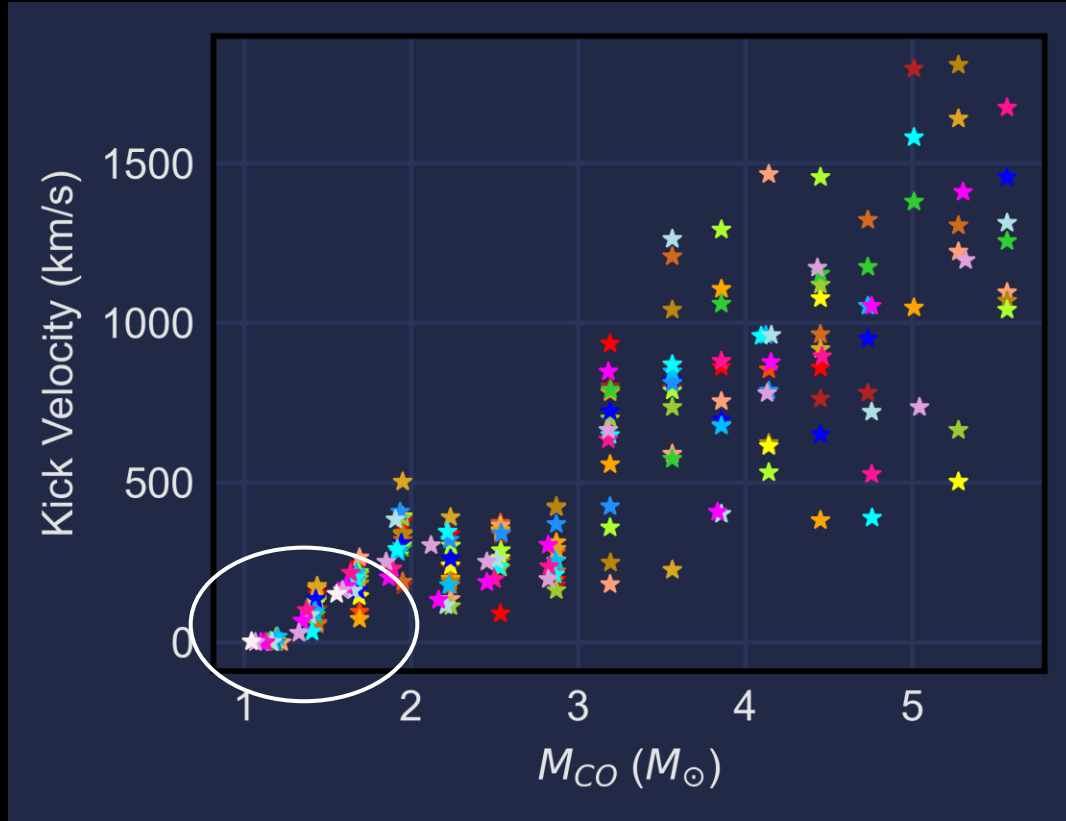
Modified Model



Remnant masses (NS2) as a function of CO core mass

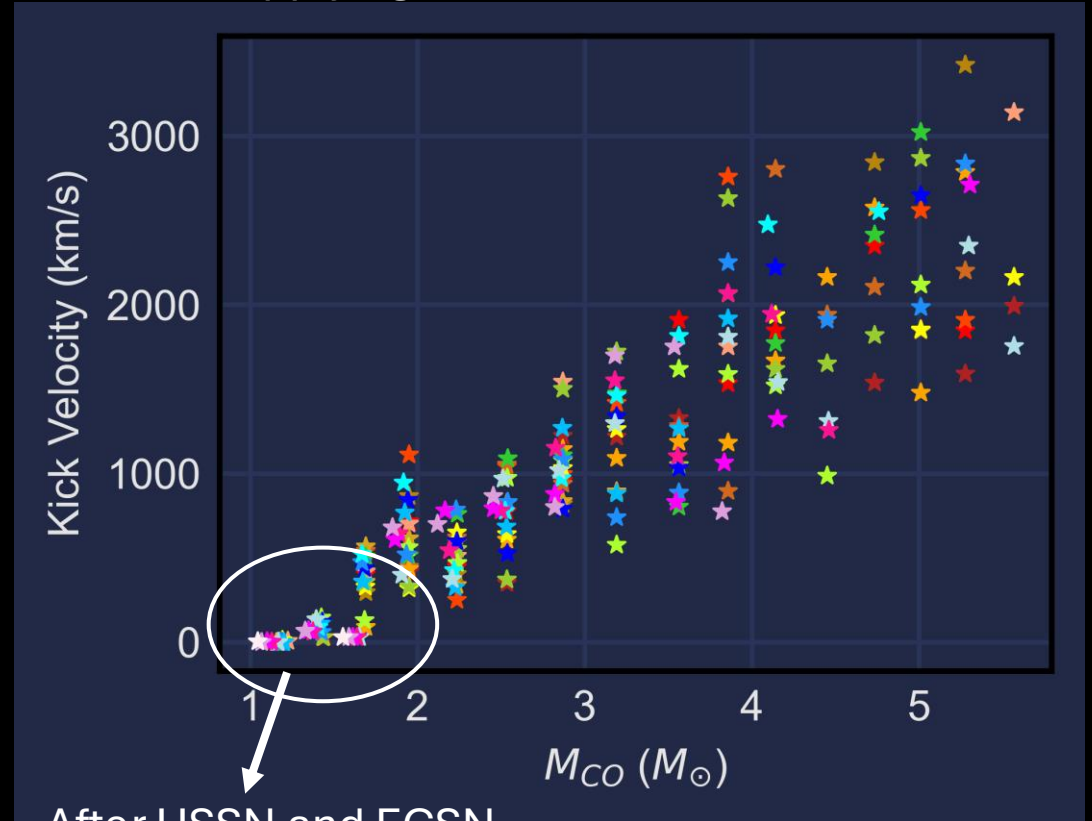
Standard Model

Before applying USSN and ECSN condition



Modified Model

After applying USSN and ECSN condition



Kick velocity for the corresponding NS2 as a function of CO core mass